Improving the Performance of Cell-Edge Users in MISO-NOMA Systems Using TAS and SWIPT-based Cooperative Transmissions

Tri Nhu Do, *Student Member, IEEE,* Daniel Benevides da Costa, *Senior Member, IEEE,* Trung Q. Duong, *Senior Member, IEEE,* and Beongku An, *Member, IEEE.*

***Abstract*—In this paper, we study the performance of a cell-edge user in a two-user multiple-input single-output non-orthogonal multiple access (MISO-NOMA) system. Since the outage performance and fairness data rate of cell-edge users are essential issues in NOMA systems, we focus on how to resolve such problems. To this end, we propose three cooperative downlink transmission schemes utilizing hybrid simultaneous wireless information and power transfer (SWIPT) and transmit antenna selection (TAS) protocols. Particularly, in each scheme, a cell-center user acts as a relay to assist the cell-edge user and its relaying operation is powered by a hybrid time-switching/power-splitting (TS/PS) SWIPT protocol. Additionally, each scheme employs a different TAS criterion to exploit the spatial diversity gain of a multiple antennas base station. In particular, we derive tight closed-form approximate expressions for the outage probabilities (OPs) and the corresponding asymptotic OPs to provide significant insights into the impact of our proposed schemes on the system performance. Our numerical results demonstrate the achievable performance improvements of the proposed schemes in comparison to that of the orthogonal multiple access (OMA) and non-cooperative NOMA systems. In addition, the proposed schemes achieve various level of diversity gains accompanying with different complexity requirements.**

***Index Terms*—Cooperative relaying transmissions, multiple-input single-output (MISO), non-orthogonal multiple access (NOMA), outage performance, simultaneous wireless information and power transfer (SWIPT), transmit antenna selection (TAS)**

The work of T. N. Do and B. An was supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (Grant No. 2016R1D1A1B03934898) and by the Leading Human Resource Training Program of Regional Neo industry Through the National Research Foundation of Korea (NRF) funded by the Ministry of Science, ICT and future planning (Grant No. 2016H1D5A1910577). The work of T. Q. Duong was supported in part by the U.K. Royal Academy of Engineering Research Fellowship under Grant RF1415\14\22 and by the U.K. Engineering and Physical Sciences Research Council (EPSRC) under Grant EP/P019374/1. The work of D. B. da Costa was supported by the National Council of Scientic and Technological Development (CNPq) under Grant 304301/2014-0. This paper was presented in part at the IEEE International Conference on Communications, Paris, France, May 2017.

T. N. Do is with the Department of Electronics and Computer Engineering in Graduate School, Hongik University, Sejong 30016, Republic of Korea (email: dotrinhu@gmail.com).

D. B. da Costa is with the Department of Computer Engineering, Federal University of Ceará, Sobral, CE 62010-560, Brazil (email: danielbcosta@ieee.org).

T. Q. Duong is with the Department of Electronics, Electrical Engineering and Computer Science, Queen's University Belfast, Belfast BT7 1NN, U.K. (e-mail: trung.q.duong@qub.ac.uk).

B. An is with the Department of Computer and Information Communications Engineering, Hongik University, Sejong 30016, Republic of Korea (email: beongku@hongik.ac.kr).

I. Introduction

Non-orthogonal multiple access (NOMA) has been emerging as a promising solution to improving spectral efficiency for the next generation of wireless communication systems [1]–[4]. The underlying concept of NOMA is that user multiplexing is conducted in the power domain [1], which is different from conventional orthogonal multiple access (OMA) techniques (e.g., time/frequency/code division multiple access). For example, in a two-user NOMA system, a base station (BS) communicates concurrently with the two users, one of which has a stronger channel condition (which is often located near the BS or at the cell-center) and the other has much weaker channel condition (which is often a cell-edge user). At the BS, information signals of the two users are superposed with different power allocations, where the power allocation coefficient of the cell-edge user is higher than that of the cell-center user. At the receiver side, the superposed signal is separated using a successive interference cancellation (SIC) technique [1]. As reported in [4], NOMA can achieve a gain over OMA, and improve cell-edge user throughput by 34.2% under a specific setting.

A. Motivation and Related Works

By using NOMA, the spectral efficiency can be significantly improved since both cell-center and cell-edge users are scheduled together and benefit from being assigned more bandwidth [2]. However, recent works in the literature have reported that the operation of NOMA raises the user fairness issue of data rate between the cell-center and cell-edge users. Indeed, the user throughput fairness is a critical issue as the rate of the cell-edge users is often lower than that of the cell-center users [5]. Additionally, as reported in [6], the power allocation coefficient of the cell-center user has to be close to zero if the cell-edge user needs a data rate comparable to that of the cell-center user. Hence, it may harm the quality-of-service (QoS) of the cell-center users since a major part of the power budget is allocated to cell-edge users, otherwise, it may compromise the reception reliability of the cell-edge users [7].

Thus, how to not only resolving the user throughput fairness but also improving the reception reliability of the cell-edge user is an important issue in NOMA context. One possible solution to avoid such fairness issue, yet guarantee the performance and reliability of the cell-edge user is to use cooperative relaying transmissions. The authors in [8] proposed a

cooperative NOMA transmission scheme, in which cell-center users (with better channel conditions) exploit prior information available in NOMA system to help improve the reception reliability of cell-edge users (with poor connections to a BS). The results from [8] also demonstrated that cooperative NOMA transmissions provide superior performance in terms of outage probability (OP) than conventional OMA and non-cooperative NOMA systems. In [9], the authors showed that using cooperative relaying transmissions where the cell-center user acts as a relay, the sum-rate of NOMA networks can be significantly improved. However the performance gap between cell-center and cell-edge users is still remaining large. In [10], the authors considered a two-user NOMA network, where the best cell-center user is selected to serve as a relay to help improve the outage performance of a selected cell-edge user.

While cooperative relaying transmissions is a sustainable solution to resolve the mentioned issues in NOMA systems, the raised question is how the cell-center users fairly consume their energy since they need to process their own data and forward the cell-edge users' data. Simultaneous wireless information and power transfer (SWIPT) is an appropriate answer since cell-center users can scavenge energy from a BS's signal and use this harvested energy to power their relaying operation. In [11], the authors combined the cooperative NOMA with SWIPT to propose a new spectral and energy efficient wireless multiple access protocol, namely, the cooperative SWIPT NOMA protocol. Specifically, the authors in [11] showed that the use of SWIPT not only is able to self-power the cell-center users but also does not jeopardize the diversity gain of the cell-edge users compared to the conventional cooperative NOMA. In [12], the authors investigated the performance of SWIPT networks where the downlink scenario employs two communication protocols, i.e., time division multiple access (TDMA) and NOMA, while the uplink scenario considers NOMA with time-sharing.

Applications of NOMA to multiple-input single-output (MISO) systems have also been studied in the literature [13], [14]. The authors in [13] investigated the impact of the concept of quasi-degradation on MISO-NOMA downlink transmission. In addition, considering a two-user MISO-NOMA system given a pair of target interference levels, a quality-of-service (QoS) optimization problem was studied in [14]. It is noteworthy that transmit antenna selection (TAS) has been demonstrated as a low-complexity and power-efficient communication scheme for the BS with multiple antennas [15], [16]. More specifically, TAS schemes can be considered as a good tradeoff between the diversity gain and the implementation cost [17]. In [18], the authors developed antenna selection and user scheduling algorithms to maximize the sum-rate of multiple-input multiple-output (MIMO)-NOMA networks, in which different selected antennas sever different pairs of users. Recently, in [19], a TAS scheme for NOMA downlink energy harvesting (EH) multiple-antenna relaying networks was studied. Specifically, the authors assumed that direct links between a source and destinations are unavailable; thus, there is a dedicated EH amplify-and-forward (AF) relay to assist the communication from the source to the destination where NOMA is performed at the relay.

B. Contributions

In this paper, we consider a two-user cooperative MISO-NOMA system, where a cell-center user acts as a relay to assist the communication from a BS to a cell-edge user. Our methods not only improve the performance of the cell-edge user but also achieve energy efficiency[1] for the cell-center user. To this end, we propose three cooperative transmission schemes in which, different TAS criteria are employed at the BS, SWIPT and decode-and-forward (DF) relaying strategy are employed at the cell-center user, and selection combining (SC) is employed at the cell-edge user. The main contributions of the paper can be summarized as follows.

- Three proposed schemes, namely Schemes I, II, and III, employ three different TAS criteria. Specifically, Scheme I aims to achieve an optimal outage performance at the cell-edge user while requires a high complexity TAS criterion. In contrast, Schemes II and III provide sub-optimal outage performance at the cell-edge user while reduce the complexity of the TAS criteria. In particular, the TAS criterion adopted in Scheme II maximizes the performance of the direct link from the BS to the cell-edge user; while that in Scheme III tries to achieve maximum amount of harvested energy, which will be used to power the relaying operation, at the cell-center user.
- For a comprehensive investigation on SWIPT protocol, we consider a generalized SWIPT architecture, i.e., the hybrid time-switching/power-splitting (TS/PS) EH receiver. We derive tight closed-form approximate expressions for the OPs of the cell-edge user and the corresponding asymptotic OPs achieved by the three proposed schemes.
- We show that the proposed schemes significantly improve the cell-edge outage performance compared to the conventional OMA and the non-cooperative NOMA systems. In particular, Scheme I is superior than the others but requires highest complexity order. Scheme III outperforms Scheme II at low signal-to-noise ratio (SNR) regime and vice versa at high SNR regime. Interestingly, Scheme I and Scheme II achieve the same diversity gain, i.e., $K+1$, while that achieved by Scheme III is always 2, where K denotes the number of antennas at the BS.

C. Structure

The rest of the paper is arranged as follows. Section II introduces the system model and describes in detail the proposed schemes. Section III presents the performance investigations in terms of the OP for the considered scenarios, and the proofs

[1]The meaning of energy efficiency considered in this paper can be explained as follows. In order to assist a cell-edge user, a cell-center user needs to spend its own energy to transmit not-its-own-data, i.e., the message of the cell-edge user. To deal with this energy consumption fairness issue at the cell-center user, we implement simultaneous information and power transfer (SWIPT) so that the cell-center user can harvest energy from the source's signal and then uses this harvested energy to forward the message of the cell-edge user. By using SWIPT, we imply that the energy efficiency has been achieved since the cell-center user does not need to spend its own energy to relay the message of the cell-edge user.

of the analytical expressions are provided in the appendixes. Section IV presents the asymptotic analysis of the derived outage performance. Section V presents some illustrative numerical results, based on which insightful discussions are provided. Monte Carlo simulations are shown to corroborate the proposed analysis. Finally, Section VI concludes the paper.

II. System Model

As shown in Fig. 1, we consider a MISO-NOMA downlink transmission where a BS, denoted by S, simultaneously communicates with a cell-center user, named User N, and a cell-edge user, called User F, by employing a two-user NOMA scheme. The BS is equipped with K antennas while each user is equipped with single antenna.

Let $h_{i\mathsf{T}}$ denote the fading coefficient of a channel from an antenna $i, i = 1, \ldots, K$ to a User T, where $\mathsf{T} \in \{\mathsf{N}, \mathsf{F}\}$. Assuming all wireless channels in the network exhibit Rayleigh block flat fading, $h_{i\mathsf{T}}$ can be modeled as independent and identically distributed (i.i.d.) complex Gaussian random variables with zero-mean and variance λ_{ST}. Additionally, let n_{aT} and n_{cT} denote the additive white Gaussian noise (AWGN) at the receiving antenna and the down-converter at User T, respectively, with zero-mean and variance σ^2_{aT} and σ^2_{cT}, respectively. Thus, the channel gain $|h_{\mathsf{XY}}|^2$, where $\mathsf{X} \in \{i, \mathsf{N}\}$ and $\mathsf{Y} \in \{\mathsf{N}, \mathsf{F}\}$, is an exponential random variable with probability density function (PDF), $f_{|h_{\mathsf{XY}}|^2}(z) = \frac{1}{\lambda_{\mathsf{XY}}} e^{-\frac{z}{\lambda_{\mathsf{XY}}}}, \forall z \geq 0$, otherwise, i.e., $z < 0$, $f_{|h_{\mathsf{XY}}|^2}(z) = 0$, where λ_{XY} denotes the mean of $|h_{\mathsf{XY}}|^2$. Additionally, the average channel gain can be written as $\mathrm{E}[|h_{\mathsf{XY}}|^2] = \frac{\mathcal{L}}{(d_{\mathsf{XY}}/d_0)^{\epsilon}}$ [20], where d_{XY} represents the distance between two nodes (in meters), ϵ stands for path-loss exponent, d_0 denotes the reference distance, and $\mathcal{L}$ is the average signal power attenuation at d_0.

In the proposed network, let User N play the role of a hybrid time-switching/power-splitting (TS/PS) EH relay (for more details related to the time-switching receiver, power-slitting receiver, and hybrid TS/PS receiver, please refer to [21], [22], and references therein). In particular, a block time T is divided into three sub-blocks. In the first sub-block with αT duration time, User N first harvests energy from its received observation, where $0 \leq \alpha < 1$ denotes the fraction of block time for energy harvesting. In the second sub-block with $(1-\alpha)T/2$ duration time, User N simultaneously utilizes a fraction ρ of the received power for energy harvesting and the remaining fraction $(1-\rho)$ for information decoding, where $0 \leq \rho < 1$ denotes the power-splitting ratio. In the third sub-block with $(1-\alpha)T/2$ duration time, User N employs all harvested energy to power its relaying operation. Fig. 2 shows an illustration of the time structure of the hybrid TS/PS SWIPT-based cooperative relaying transmission.

Note that we consider the hybrid TS/PS SWIPT to provide a comprehensive investigation and develop a general analytical framework for SWIPT architecture. Also, the hybrid receiver provides a flexible choice for system designing. For example, in the case of TS (or PS) is not necessary, α (or ρ) is simply set to zero. The order of which module, i.e., TS or PS, is performed first may not be important since their roles are equal [21]. Additionally, by considering TS first, the time structure

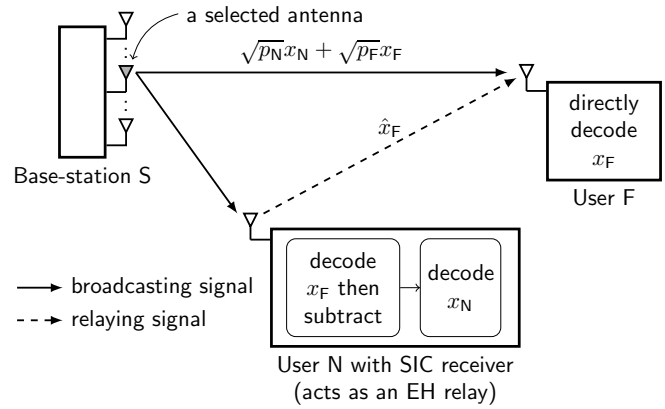

Fig. 1. An illustration of a hybrid time-switching/power-splitting (TS/PS) SWIPT-based cooperative relaying transmission with TAS for a two-user MISO-NOMA system.

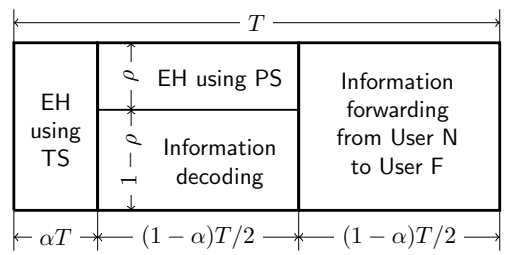

Fig. 2. Illustration of a hybrid TS/PS SWIPT-based DF relaying strategy.

as depicted in Fig. 2 is more reasonable since the information decoding and relaying processes are consecutive.

According to the time frame of the hybrid TS/PS EH protocol as shown in Fig. 2, the downlink scenario of the MISO-NOMA cooperative relaying transmission is conducted in two phases, namely the energy harvesting and direct information transmission phase, which is carried out in the first and second sub-blocks, and the cooperative relaying transmission phase, which is carried out in the third sub-block. Note that the durations of the second and third sub-blocks are equal. In addition, the BS keeps silent[2] while User N transmits to User F during the third sub-block, i.e., during the cooperative relaying transmission phase.

A. The First Phase: Energy Harvesting and Direct Information Transmission

Suppose that antenna i on the BS has been selected for information transmission. According to the principle of NOMA, the intended transmit messages x_{N} and x_{F} of Users N and F, respectively, are superposed as $\sqrt{p_{\mathsf{N}}}x_{\mathsf{N}} + \sqrt{p_{\mathsf{F}}}x_{\mathsf{F}}$ and then broadcasted by the selected antenna at the beginning of the first

[2]Note that both direct and relaying transmissions are performed on the same frequency. Thus, if the BS still transmits while User N forwards messages to User F, the cell-edge user will suffer from co-channel interference (CCI) created by the transmission of the BS. Therefore, in order to avoid such CCI, the direct and relaying transmissions are carried out in two separated sub-blocks. Please note that the use of such two separated sub-blocks, in which the BS keeps silent when a relay transmits to a destination, has been well-adopted in the literature, as in [8], [23]

sub-block time, where p_{N} and p_{F} denote the power allocation coefficients (PACs) for Users N and F, respectively. Following the principle of NOMA, we assume that $|h_{i\mathsf{N}}|^2 > |h_{i\mathsf{F}}|^2$, $0 < p_{\mathsf{N}} < p_{\mathsf{F}}$, and $p_{\mathsf{N}} + p_{\mathsf{F}} = 1$ [1].

1) At User N*:* The received observation the antenna of User N can be written as

$$y_{i\mathsf{N}} = \left(\sqrt{p_{\mathsf{N}} P_{\mathsf{S}}} x_{\mathsf{N}} + \sqrt{p_{\mathsf{F}} P_{\mathsf{S}}} x_{\mathsf{F}}\right) h_{i\mathsf{N}} + n_{\mathsf{aN}}, \tag{1}$$

where where $h_{i\mathsf{N}} \sim \mathcal{CN}(0, \lambda_{\mathsf{SN}})$, $n_{\mathsf{aN}} \sim \mathcal{CN}(0, \sigma^2_{\mathsf{aN}})$. Using the hybrid TS/PS EH protocol, the total harvested energy at User N associated with antenna i can be expressed as

$$E_{i\mathsf{N}} = \eta P_{\mathsf{S}} |h_{i\mathsf{N}}|^2 \alpha T + \eta \rho P_{\mathsf{S}} |h_{i\mathsf{N}}|^2 (1-\alpha) T/2, \tag{2}$$

where $0 < \eta < 1$ denotes the energy conversion efficiency, $|h_{i\mathsf{N}}|^2$ denotes the channel gain between antenna i and User N. For information decoding (ID), the received signal at User N can be written as

$$y^{\mathtt{ID}}_{i\mathsf{N}} = \sqrt{1-\rho}\left[\left(\sqrt{p_{\mathsf{N}} P_{\mathsf{S}}} x_{\mathsf{N}} + \sqrt{p_{\mathsf{F}} P_{\mathsf{S}}} x_{\mathsf{F}}\right) h_{i\mathsf{N}} + n_{\mathsf{aN}}\right] + n_{\mathsf{cN}}, \tag{3}$$

where $n_{\mathsf{cN}} \sim \mathcal{CN}(0, \sigma^2_{\mathsf{cN}})$.

According to the principle of NOMA, the SIC receiver at User N first decodes x_{F} and then subtracts this component from the received signal to detect its own message, i.e., x_{N} [1]. Thus, the received signal-to-interference-plus-noise ratio (SINR) at User N to decode x_{F} can be expressed as

$$\gamma^{x_{\mathsf{F}}}_{i\mathsf{N}} = \frac{(1-\rho) p_{\mathsf{F}} P_{\mathsf{S}} |h_{i\mathsf{N}}|^2}{(1-\rho) p_{\mathsf{N}} P_{\mathsf{S}} |h_{i\mathsf{N}}|^2 + (1-\rho)\sigma^2_{\mathsf{aN}} + \sigma^2_{\mathsf{cN}}}, \tag{4}$$

and the received signal-to-noise ratio (SNR) at User N to decode x_{N} can be written as

$$\gamma^{x_{\mathsf{N}}}_{i\mathsf{N}} = \frac{(1-\rho) p_{\mathsf{N}} P_{\mathsf{S}} |h_{i\mathsf{N}}|^2}{(1-\rho)\sigma^2_{\mathsf{aN}} + \sigma^2_{\mathsf{cN}}}. \tag{5}$$

2) At User F*:* The received observation at User F can be expressed as[3]

$$y_{i\mathsf{F}} = \left(\sqrt{p_{\mathsf{N}} P_{\mathsf{S}}} x_{\mathsf{N}} + \sqrt{p_{\mathsf{F}} P_{\mathsf{S}}} x_{\mathsf{F}}\right) h_{i\mathsf{F}} + n_{\mathsf{aF}} + n_{\mathsf{cF}}, \tag{6}$$

where $h_{i\mathsf{F}} \sim \mathcal{CN}(0, \lambda_{\mathsf{SF}})$, $n_{\mathsf{aF}} \sim \mathcal{CN}(0, \sigma^2_{\mathsf{aF}})$, and $n_{\mathsf{cF}} \sim \mathcal{CN}(0, \sigma^2_{\mathsf{cF}})$.

In contrast with User N, User F can directly decode its information signal since User F is allocated with higher transmit power and thus the interference introduced by the information signal of User N can be considered as noise [1]. Thus, the received SNR at User F to decode x_{F} can be written as

$$\gamma_{i\mathsf{F}} = \frac{p_{\mathsf{F}} P_{\mathsf{S}} |h_{i\mathsf{F}}|^2}{p_{\mathsf{N}} P_{\mathsf{S}} |h_{i\mathsf{F}}|^2 + \sigma^2_{\mathsf{aF}} + \sigma^2_{\mathsf{cF}}}. \tag{7}$$

B. The Second Phase: Cooperative Relaying Transmission

Assuming that all the energy harvested during the first phase is employed to power the relaying operation, as in [21], [22], the transmit power of User N in the second phase can be expressed as

$$P_{\mathsf{N}} = \frac{E_{i\mathsf{N}}}{(1-\alpha)T/2} = \eta P_{\mathsf{S}} |h_{i\mathsf{N}}|^2 \left(\frac{2\alpha}{1-\alpha} + \rho\right). \tag{8}$$

[3]Note that in the first sub-block, User F can perform EH or keep silent, and in the second sub-block, it only performs information decoding.

Considering the DF relaying protocol, the received signal at User F can be written as

$$y_{\mathsf{NF}} = \sqrt{P_{\mathsf{N}}} h_{\mathsf{NF}} \hat{x}_{\mathsf{F}} + n_{\mathsf{aF}} + n_{\mathsf{cF}}, \tag{9}$$

where $h_{\mathsf{NF}} \sim \mathcal{CN}(0, \lambda_{\mathsf{NF}})$, and $\hat{x}_{\mathsf{F}}$ denotes the re-encoded version of x_{F}.

From (8) and (9), the received SNR at User F to detect x_{F} transmitted by User N can be written as

$$\gamma_{\mathsf{NF}} = \frac{\eta P_{\mathsf{S}} |h_{i\mathsf{N}}|^2 |h_{\mathsf{NF}}|^2 \left(\frac{2\alpha}{1-\alpha} + \rho\right)}{\sigma^2_{\mathsf{aF}} + \sigma^2_{\mathsf{cF}}}. \tag{10}$$

Finally, User F combines two signals, i.e., the direct signal from the BS and the relaying signal from User N by employing a selection combining (SC) technique. Thus, the achievable SNR of the combination of two received signals at User F can be expressed as

$$\gamma^{\mathtt{SC}}_{\mathsf{F}} = \max\{\gamma_{i\mathsf{F}}, \gamma_{\mathsf{NF}}\}. \tag{11}$$

C. The Proposed Transmit Antenna Selection (TAS) Criteria

The proposed TAS schemes are conducted before data transmission through the signaling and channel state information (CSI) estimation/calculation system. We assume that the required CSI of each scheme is available [8], [11]. Let i^* denote the selected antenna, the three proposed TAS schemes are described as follows.

1) The TAS Criterion Adopted in Scheme I: It can be observed that the cooperative relaying operation depends on whether x_{F} can be decoded at User N or not. Thus, the end-to-end SNR at User F can be written as

$$\gamma^{\mathtt{e2e}}_{\mathsf{F}} = \min\{\gamma^{x_{\mathsf{F}}}_{i\mathsf{N}}, \gamma^{\mathtt{SC}}_{\mathsf{F}}\}. \tag{12}$$

The instantaneous transmission rate achieved by User F associating with antenna i can be expressed as

$$R_{i\mathsf{F}} = \frac{1-\alpha}{2} \log_2(1 + \gamma^{\mathtt{e2e}}_{\mathsf{F}}). \tag{13}$$

The TAS criterion adopted in Scheme I aims to select an antenna that maximizes the instantaneous transmission rate of User F. Mathematically, the selection Scheme I can be expressed as

$$\begin{aligned} i^* &= \frac{1-\alpha}{2} \log_2(1 + \gamma^{\mathtt{e2e}}_{\mathsf{F}}) \\ &= \arg \max_{1 \le i \le K} \min\{\gamma^{x_{\mathsf{F}}}_{i\mathsf{N}}, \max\{\gamma_{i\mathsf{F}}, \gamma_{\mathsf{NF}}\}\}. \end{aligned} \tag{14}$$

2) The TAS Criterion Adopted in Scheme II: As can be observed, with Scheme I, User F can achieve optimal outage performance. However, Scheme I requires a high complexity operation, i.e., CSI of three kinds of channels, namely $\mathsf{S} \to \mathsf{N}, \mathsf{S} \to \mathsf{F}$, and $\mathsf{N} \to \mathsf{F}$ channels is needed. In order to reduce the complexity of Scheme I but also achieve a comparable outage performance, we propose Scheme II by focusing on the performance of User F during the first phase, i.e., the broadcasting phase.

Note that, during the broadcasting phase, the message x_{F} of User F needs to be decoded not only at User F but also at the SIC receiver of User N. Thus, the instantaneous transmission

$$R_{i\mathsf{F}}^{(1)} = \min\left\{\frac{1-\alpha}{2}\log_2(1+\gamma_{i\mathsf{N}}^{x_\mathsf{F}}), \frac{1-\alpha}{2}\log_2(1+\gamma_{i\mathsf{F}})\right\}$$
$$= \frac{1-\alpha}{2}\log_2\left(1+\min\left\{\frac{(1-\rho)p_\mathsf{F}P_\mathsf{S}|h_{i\mathsf{N}}|^2}{(1-\rho)p_\mathsf{N}P_\mathsf{S}|h_{i\mathsf{N}}|^2+(1-\rho)\sigma_{\mathsf{aN}}^2+\sigma_{\mathsf{cN}}^2}, \frac{p_\mathsf{F}P_\mathsf{S}|h_{i\mathsf{F}}|^2}{p_\mathsf{N}P_\mathsf{S}|h_{i\mathsf{F}}|^2+\sigma_{\mathsf{aF}}^2+\sigma_{\mathsf{cF}}^2}\right\}\right) \quad (15)$$

rate achieved by User F associating with antenna i can be expressed as on the top of the next page, where the superscript (1) indicates the first phase. Since User N is located nearer the BS than User F, thus, $|h_{i\mathsf{N}}|^2 \gg |h_{i\mathsf{F}}|^2$. Also, the conversion noise power is significantly less than the received power. Hence, $R_{i\mathsf{F}}$ can be written as

$$R_{i\mathsf{F}}^{(1)} = \frac{1-\alpha}{2}\log_2\left(1+\frac{p_\mathsf{F}P_\mathsf{S}|h_{i\mathsf{F}}|^2}{p_\mathsf{N}P_\mathsf{S}|h_{i\mathsf{F}}|^2+\sigma_{\mathsf{aF}}^2+\sigma_{\mathsf{cF}}^2}\right). \quad (16)$$

The TAS criterion adopted in Scheme II aims to select an antenna that maximizes $R_{i\mathsf{F}}^{(1)}$, but offers relatively low computational complexity, which can be mathematically expressed as

$$i^* = \arg\max_{1\le i\le K}\frac{1-\alpha}{2}\log_2\left(1+\frac{p_\mathsf{F}P_\mathsf{S}|h_{i\mathsf{F}}|^2}{p_\mathsf{N}P_\mathsf{S}|h_{i\mathsf{F}}|^2+\sigma_{\mathsf{aF}}^2+\sigma_{\mathsf{cF}}^2}\right)$$
$$= \arg\max_{1\le i\le K}\frac{p_\mathsf{F}P_\mathsf{S}|h_{i\mathsf{F}}|^2}{p_\mathsf{N}P_\mathsf{S}|h_{i\mathsf{F}}|^2+\sigma_{\mathsf{aF}}^2+\sigma_{\mathsf{cF}}^2}$$
$$= \arg\max_{1\le i\le K}|h_{i\mathsf{F}}|^2. \quad (17)$$

3) The TAS Criterion Adopted in Scheme III: We now turn our attention into the amount of harvested energy at the cell-center user, i.e., $E_{i\mathsf{N}}$ in (2). Note that all the harvested energy will be used by User N to transmit x_F to User F. The TAS criterion adopted in Scheme III selects an antenna that provides a maximum instantaneous harvested energy at User N. Mathematically, the selection Scheme III can be expressed as

$$i^* = \arg\max_{1\le i\le K}\left\{\eta P_\mathsf{S}|h_{i\mathsf{N}}|^2\alpha T + \eta\rho P_\mathsf{S}|h_{i\mathsf{N}}|^2(1-\alpha)T/2\right\}$$
$$= \arg\max_{1\le i\le K}|h_{i\mathsf{N}}|^2. \quad (18)$$

The performance investigation of the three proposed schemes in terms of OP and asymptotic OP will be presented in the next section.

III. Outage Performance Analysis

The OP of a user can be defined as the probability that the instantaneous data rate of the user falls below a predefined target data rate [24].

Let $R_{\mathtt{th},\mathsf{N}}$ and $R_{\mathtt{th},\mathsf{F}}$ (bits/s/Hz) denote the target data rates of Users N and F, respectively; $a_1 \triangleq \frac{(1-\rho)p_\mathsf{F}P_\mathsf{S}}{(1-\rho)\sigma_{\mathsf{aN}}^2+\sigma_{\mathsf{cN}}^2}$, $a_2 \triangleq \frac{(1-\rho)p_\mathsf{N}P_\mathsf{S}}{(1-\rho)\sigma_{\mathsf{aN}}^2+\sigma_{\mathsf{cN}}^2}$, $b_1 \triangleq \frac{p_\mathsf{F}P_\mathsf{S}}{\sigma_{\mathsf{aF}}^2+\sigma_{\mathsf{cF}}^2}$, $b_2 \triangleq \frac{p_\mathsf{N}P_\mathsf{S}}{\sigma_{\mathsf{aF}}^2+\sigma_{\mathsf{cF}}^2}$, and $c \triangleq \eta P_\mathsf{S}\left(\frac{2\alpha}{1-\alpha}+\rho\right)/(\sigma_{\mathsf{aF}}^2+\sigma_{\mathsf{cF}}^2)$, $\mu_a \triangleq \frac{\gamma_2}{a_1-a_2\gamma_2}$, $\mu_b \triangleq \frac{\gamma_2}{b_1-b_2\gamma_2}$, and $\theta \triangleq \frac{p_\mathsf{F}}{p_\mathsf{N}}$. Function $\Omega(\mu,\chi,\xi)$ is defined in (21). For the sake of notational convenience, let $X_i \triangleq |h_{i\mathsf{N}}|^2, Y \triangleq |h_{\mathsf{NF}}|^2$, and $Z_i \triangleq |h_{i\mathsf{F}}|^2$.

A. Outage Performance Analysis of Scheme I

The OP of User F in Scheme I can be expressed as

$$P_{\mathrm{out},\mathsf{F}}^{(\mathrm{I})} = \Pr(R_{i\mathsf{F}} < R_{\mathtt{th},\mathsf{F}})$$
$$= \Pr\left(\max_{1\le i\le K}\min\{\gamma_{i\mathsf{N}}^{x_\mathsf{F}}, \max\{\gamma_{i\mathsf{F}},\gamma_{\mathsf{NF}}\}\} < \gamma_2\right)$$
$$= \Pr\left(\max_{1\le i\le K}\min\left\{\frac{a_1X_i}{a_2X_i+1}, \max\left\{\frac{b_1Z_i}{b_2Z_i+1}, cX_iY\right\}\right\} < \gamma_2\right), \quad (19)$$

where $\gamma_2 \triangleq 2^{2R_{\mathtt{th},\mathsf{F}}/(1-\alpha)}-1$ denotes the SNR threshold for correctly decoding the message x_F.

Theorem 1: The closed-form approximate expression for the OP of User F in Scheme I can be attained as

$$P_{\mathrm{out},\mathsf{F}}^{(\mathrm{I})} = \sum_{i,j,k}^{K=i+j+k}\binom{K}{i,j,k}\left(1-e^{-\frac{\mu_a}{\lambda_{\mathsf{SN}}}-\frac{\mu_b}{\lambda_{\mathsf{SF}}}}\right)^i(-1)^j$$
$$\times e^{-\frac{k\mu_b}{\lambda_{\mathsf{SF}}}}\frac{1}{\lambda_{\mathsf{NF}}}\left[\sqrt{\frac{4(j+k)\lambda_{\mathsf{NF}}\gamma_2}{\lambda_{\mathsf{SN}}c}}K_1\left(\sqrt{\frac{4(j+k)\gamma_2}{\lambda_{\mathsf{SN}}\lambda_{\mathsf{NF}}c}}\right)\right.$$
$$\left.-\Omega\left(\frac{\gamma_2}{c\mu_a},\frac{1}{\lambda_{\mathsf{NF}}},\frac{(j+k)\gamma_2}{\lambda_{\mathsf{SN}}c}\right)\right]$$
$$+\left(1-e^{-\frac{\mu_a}{\lambda_{\mathsf{SN}}}}\right)^K e^{-\frac{\gamma_2}{\lambda_{\mathsf{NF}}c\mu_a}}, \quad (20)$$

where

$$\Omega(\mu,\chi,\xi)$$
$$= \frac{e^{-\mu\chi}}{\chi} - \xi\Gamma(0,\mu\chi) + \sum_{u=2}^{\infty}\frac{(-1)^u\xi^u}{u!}$$
$$\times\left[e^{-\mu\chi}\sum_{v=1}^{u-1}\frac{(v-1)!(-\chi)^{u-v-1}}{(u-1)!\mu^v} - \frac{(-\chi)^{u-1}}{(u-1)!}\mathrm{Ei}(-\mu\chi)\right], \quad (21)$$

where $\Gamma(\cdot,\cdot)$ is the upper incomplete Gamma function [25, Eq. (8.350.2)], $\mathrm{Ei}(\cdot)$ is the exponential integral function [25, Eq. (8.211.1)].

Proof: See Appendix A. ■

B. Outage Performance Analysis of Scheme II

The achievable end-to-end transmission rate at User F can be expressed in another form as [26]

$$R_{i\mathsf{F}}^{\mathsf{e2e}} = \begin{cases}\frac{1-\alpha}{2}\log_2(1+\gamma_{i\mathsf{F}}), & \text{if } R_{i\mathsf{N}}^{x_\mathsf{F}} < R_{\mathtt{th},\mathsf{F}},\\ \frac{1-\alpha}{2}\log_2(1+\gamma_\mathsf{F}^{\mathsf{SC}}), & \text{if } R_{i\mathsf{N}}^{x_\mathsf{F}} \ge R_{\mathtt{th},\mathsf{F}},\end{cases} \quad (22)$$

where $R_{i\mathsf{N}}^{x_\mathsf{F}} = \frac{1-\alpha}{2}\log_2(1+\gamma_{i\mathsf{N}}^{x_\mathsf{F}})$. The factor $\frac{1-\alpha}{2}$ appears because the direct information transmission or the cooperative

relaying transmission is conducted during $\frac{(1-\alpha)T}{2}$ duration time [21], [26].

Theorem 2: The closed-form approximate expression for the OP of User F in Scheme II can be written as

$$P_{\text{out,F}}^{(\text{II})} = \sum_{k=0}^{K} \binom{K}{k} (-1)^k e^{-\frac{k\mu_b}{\lambda_{\text{SF}}}} \left[1 - \frac{1}{\lambda_{\text{SN}}} \Omega\left(\mu_a, \frac{1}{\lambda_{\text{SN}}}, \frac{\gamma_2}{\lambda_{\text{NF}} c}\right) \right], \tag{23}$$

if $\gamma_2 < \theta$, o.w., $P_{\text{out,F}}^{(\text{II})} = 1$.

Proof: See Appendix B. ■

C. Outage Performance Analysis of Scheme III

Theorem 3: The closed-form approximate expression for the OP of User F in Scheme III can be written as

$$\begin{aligned} P_{\text{out,F}}^{(\text{III})} &= \left(1 - e^{-\frac{\mu_b}{\lambda_{\text{SF}}}}\right) \Bigg[\sum_{k=0}^{K} \binom{K}{k} (-1)^k e^{-\frac{k\mu_a}{\lambda_{\text{SN}}}} \\ &\quad + \sum_{l=1}^{K} \binom{K}{l} (-1)^{l+1} \left(e^{-\frac{l\mu_a}{\lambda_{\text{SN}}}} - \frac{l}{\lambda_{\text{SN}}} \Omega\left(\mu_a, \frac{l}{\lambda_{\text{SN}}}, \frac{\gamma_2}{\lambda_{\text{NF}} c}\right) \right) \Bigg], \end{aligned} \tag{24}$$

if $\gamma_2 < \theta$, o.w., $P_{\text{out,F}}^{(\text{III})} = 1$.

Proof: By following the similar calculation steps as presented in Appendix B, $P_{\text{out,F}}^{(\text{III})}$ can be obtained. This completes the proof of Theorem 3. ■

IV. Asymptotic Outage Performance Analysis

For the sake of notational convenience, let $\bar{\gamma}_{\text{N}} = \frac{P_{\text{S}}}{(1-\rho)\sigma_{\text{aN}}^2 + \sigma_{\text{cN}}^2}$, and $\bar{\gamma}_{\text{F}} = \frac{P_{\text{S}}}{\sigma_{\text{aF}}^2 + \sigma_{\text{cF}}^2}$. It can be seen that $\bar{\gamma}_{\text{N}} \approx \bar{\gamma}_{\text{F}}$ when $P_{\text{S}} \to \infty$, therefore, let $\bar{\gamma}$ represent both $\bar{\gamma}_{\text{N}}$ and $\bar{\gamma}_{\text{F}}$ at the high SNR regime. Thus, $\mu_a = \frac{\tilde{\mu}_a}{\bar{\gamma}}$, $\mu_b = \frac{\tilde{\mu}_b}{\bar{\gamma}}$, and $c = \bar{\gamma}\tilde{c}$, where $\tilde{\mu}_a = \frac{\gamma_2}{(1-\rho)(p_{\text{F}} - p_{\text{N}}\gamma_2)}$, $\tilde{\mu}_b = \frac{\gamma_2}{p_{\text{F}} - p_{\text{N}}\gamma_2}$, and $\tilde{c} = \eta\left(\frac{2\alpha}{1-\alpha} + \rho\right)$, respectively.

A. The Asymptotic OP of User F in Scheme I

Using the fact that $e^{-a/x} \approx 1 - a/x$ when $x \to \infty$, the asymptotic OP of User F in Scheme I, $P_{\text{asym,F}}^{(\text{I})}$, can be written as

$$\begin{aligned} P_{\text{asym,F}}^{(\text{I})} &= \int_0^{\frac{t}{c\mu_a}} \left[\frac{1}{\bar{\gamma}} \left(\frac{\tilde{\mu}_a}{\lambda_{\text{SN}}} + \frac{\tilde{\mu}_b}{\lambda_{\text{SF}}} \right) - \frac{1}{\bar{\gamma}^2} \frac{\tilde{\mu}_a \tilde{\mu}_b}{\lambda_{\text{SN}} \lambda_{\text{SF}}} \right. \\ &\quad \left. - \frac{1}{\bar{\gamma}} \frac{\tilde{\mu}_b}{\lambda_{\text{SF}}} e^{-\frac{t}{\lambda_{\text{SN}} c y}} \right]^K \frac{1}{\lambda_Y} e^{-\frac{y}{\lambda_Y}} dy \\ &\quad + \frac{1}{\bar{\gamma}^K} \left(\frac{\tilde{\mu}_a}{\lambda_{\text{SN}}} \right)^K e^{-\frac{\gamma_2}{\lambda_{\text{NF}} \tilde{c} \tilde{\mu}_a}}. \end{aligned} \tag{25}$$

Relying on trinomial coefficient, the integral in $P_{\text{asym,F}}^{(\text{I})}$, named $I_{\text{asym,F}}^{(\text{I})}$, can be further expressed as

$$\begin{aligned} I_{\text{asym,F}}^{(\text{I})} &= \sum_{i,j,k}^{K=i+j+k} \binom{K}{i,j,k} (-1)^{j+k} \frac{1}{\bar{\gamma}^{i+2j+k}} \\ &\quad \times \left(\frac{\tilde{\mu}_a}{\lambda_{\text{SN}}} + \frac{\tilde{\mu}_b}{\lambda_{\text{SF}}} \right)^i \left(\frac{\tilde{\mu}_a}{\lambda_{\text{SN}}} \right)^j \left(\frac{\tilde{\mu}_b}{\lambda_{\text{SF}}} \right)^{j+k} \\ &\quad \times \frac{1}{\lambda_{\text{NF}}} \int_0^{\frac{\gamma_2}{c\mu_a}} \left(e^{-\frac{y}{\lambda_{\text{NF}}}} - e^{-\frac{k\gamma_2}{\lambda_{\text{SN}} c y}} \right) dy. \end{aligned} \tag{26}$$

Using similar calculation steps as done in Appendix A, $I_{\text{asym,F}}^{(\text{I})}$ in (26) can be further expressed as in (27), which presented on the top of the next page. Relying on the approximation $xK_1(x) \approx 1 + \frac{x^2}{2} \ln \frac{x}{2}$ [11], and then substituting $I_{\text{asym,F}}^{(\text{I})}$ into (27), $P_{\text{asym,F}}^{(\text{I})}$ can be attained as in (28), which presented on the top of the next page.

B. The Asymptotic OP of User F in Scheme II

Using the fact that $e^{-a/x} \approx 1 - a/x$ when $x \to \infty$, Λ_{1A} in (57) and Λ_{1B} in (60) can be asymptotically approximated as

$$\Lambda_{1A}^{\text{asym}} = \frac{1}{\bar{\gamma}} \frac{\tilde{\mu}_a}{\lambda_{\text{SN}}}, \qquad \Lambda_{1B}^{\text{asym}} = \frac{1}{\bar{\gamma}^K} \left(\frac{\tilde{\mu}_b}{\lambda_{\text{SF}}} \right)^K. \tag{29}$$

By applying the following identities to (66), i.e., $\Gamma(0, x) = -\text{Ei}(-x)$ [25, 8.359.1] and $\text{Ei}(-x) \approx C_E + \ln(x)$ [25, Eq. (8.214.1)], where C_E denotes the Euler's constant [25, Eq. (8.367.1)], Λ_{2B} can be asymptotically approximated by

$$\begin{aligned} \Lambda_{2B}^{\text{asym}} &= -\left(\frac{\gamma_2}{\lambda_{\text{SN}} \lambda_{\text{NF}} \tilde{c} \bar{\gamma}} + \frac{\gamma_2^2}{2\lambda_{\text{SN}}^2 \lambda_{\text{NF}}^2 \tilde{c}^2 \bar{\gamma}^2} \right) \\ &\quad \times \left[C_E + \ln\left(\frac{\tilde{\mu}_a}{\lambda_{\text{SN}} \bar{\gamma}} \right) \right] \frac{\gamma_2^2}{2\lambda_{\text{SN}} \lambda_{\text{NF}}^2 \tilde{c}^2 \tilde{\mu}_a \bar{\gamma}} \left(1 - \frac{\tilde{\mu}_a}{\lambda_{\text{SN}} \bar{\gamma}} \right). \end{aligned} \tag{30}$$

Hence, by combining (29) and (30), the asymptotic OP of User F can be obtained as

$$P_{\text{asym,F}}^{(\text{II})} = \Lambda_{1B}^{\text{asym}} \left(\Lambda_{1A}^{\text{asym}} + \Lambda_{2B}^{\text{asym}} \right). \tag{31}$$

C. The Asymptotic OP of User F in Scheme III

Following the similar derivation steps as done in Subsection IV-B, the asymptotic OP of User F can be expressed as in (32), which presented on the top of the next page.

V. Numerical Results and Discussions

In this section, representative numerical results are presented to illustrate the outage performance of the three proposed schemes of the system under study. Additionally, insightful comparisons related to the achievable diversity gain are provided.

$$I_{\text{asym},\mathsf{F}}^{(\text{I})} = \sum_{i,j,k}^{K=i+j+k} \binom{K}{i,j,k} (-1)^{j+k} \frac{1}{\bar{\gamma}^{i+2j+k}} \left(\frac{\tilde{\mu}_a}{\lambda_{\mathsf{SN}}} + \frac{\tilde{\mu}_b}{\lambda_{\mathsf{SF}}}\right)^i \left(\frac{\tilde{\mu}_a}{\lambda_{\mathsf{SN}}}\right)^j \left(\frac{\tilde{\mu}_b}{\lambda_{\mathsf{SF}}}\right)^{j+k}$$
$$\times \left[\sqrt{\frac{4k\gamma_2}{\lambda_{\mathsf{SN}}\lambda_{\mathsf{NF}}\tilde{c}\bar{\gamma}}} K_1\left(\sqrt{\frac{4k\gamma_2}{\lambda_{\mathsf{SN}}\lambda_{\mathsf{NF}}\tilde{c}\bar{\gamma}}}\right) - e^{-\frac{\gamma_2}{\lambda_{\mathsf{NF}}\tilde{c}\tilde{\mu}_a}} + \frac{k\gamma_2}{\lambda_{\mathsf{SN}}\lambda_{\mathsf{NF}}\tilde{c}\bar{\gamma}} \Gamma\left(0, \frac{\gamma_2}{\lambda_{\mathsf{NF}}\tilde{c}\tilde{\mu}_a}\right)\right.$$
$$\left. - \frac{k^2\gamma_2^2}{2\lambda_{\mathsf{NF}}\lambda_{\mathsf{SN}}^2\tilde{c}^2\bar{\gamma}^2}\left(\frac{\tilde{c}\tilde{\mu}_a}{\gamma_2} e^{-\frac{\gamma_2}{\lambda_{\mathsf{NF}}\tilde{c}\tilde{\mu}_a}} + \frac{1}{\lambda_{\mathsf{NF}}} \text{Ei}\left(-\frac{\gamma_2}{\lambda_{\mathsf{NF}}\tilde{c}\tilde{\mu}_a}\right)\right)\right] \tag{27}$$

$$P_{\text{asym},\mathsf{F}}^{(\text{I})} = \sum_{i,j,k}^{K=i+j+k} \binom{K}{i,j,k} (-1)^{j+k} \frac{1}{\bar{\gamma}^{i+2j+k}} \left(\frac{\tilde{\mu}_a}{\lambda_{\mathsf{SN}}} + \frac{\tilde{\mu}_b}{\lambda_{\mathsf{SF}}}\right)^i \left(\frac{\tilde{\mu}_a}{\lambda_{\mathsf{SN}}}\right)^j \left(\frac{\tilde{\mu}_b}{\lambda_{\mathsf{SF}}}\right)^{j+k}$$
$$\times \left[1 + \frac{2k\gamma_2}{\lambda_{\mathsf{SN}}\lambda_{\mathsf{NF}}\tilde{c}\bar{\gamma}} \ln\left(\sqrt{\frac{k\gamma_2}{\lambda_{\mathsf{SN}}\lambda_{\mathsf{NF}}\tilde{c}\bar{\gamma}}}\right) - e^{-\frac{\gamma_2}{\lambda_{\mathsf{NF}}\tilde{c}\tilde{\mu}_a}} + \frac{k\gamma_2}{\lambda_{\mathsf{SN}}\lambda_{\mathsf{NF}}\tilde{c}\bar{\gamma}} \Gamma\left(0, \frac{\gamma_2}{\lambda_{\mathsf{NF}}\tilde{c}\tilde{\mu}_a}\right)\right.$$
$$\left. - \frac{k^2\gamma_2^2}{2\lambda_{\mathsf{NF}}\lambda_{\mathsf{SN}}^2\tilde{c}^2\bar{\gamma}^2}\left(\frac{\tilde{c}\tilde{\mu}_a}{\gamma_2} e^{-\frac{\gamma_2}{\lambda_{\mathsf{NF}}\tilde{c}\tilde{\mu}_a}} + \frac{1}{\lambda_{\mathsf{NF}}} \text{Ei}\left(-\frac{\gamma_2}{\lambda_{\mathsf{NF}}\tilde{c}\tilde{\mu}_a}\right)\right)\right] + \frac{1}{\bar{\gamma}^K}\left(\frac{\tilde{\mu}_a}{\lambda_{\mathsf{SN}}}\right)^K e^{-\frac{\gamma_2}{\lambda_{\mathsf{NF}}\tilde{c}\tilde{\mu}_a}} \tag{28}$$

$$P_{\text{asym},\mathsf{F}}^{(\text{III})} = \frac{1}{\bar{\gamma}} \frac{\tilde{\mu}_b}{\lambda_{\mathsf{SF}}} \left[\frac{1}{\bar{\gamma}^K}\left(\frac{\tilde{\mu}_a}{\lambda_{\mathsf{SN}}}\right)^K + \sum_{k=1}^{K} \binom{K}{k} (-1)^{k+1} k \left[\frac{\gamma_2}{\lambda_{\mathsf{SN}}\lambda_{\mathsf{NF}}\tilde{c}\bar{\gamma}}\left(-C_E - \ln\left(\frac{k\tilde{\mu}_a}{\lambda_{\mathsf{SN}}\bar{\gamma}}\right)\right)\right.\right.$$
$$\left.\left. - \frac{\gamma_2^2}{2\lambda_{\mathsf{SN}}\lambda_{\mathsf{NF}}^2\tilde{c}^2\bar{\gamma}^2}\left[\left(1 - \frac{k\tilde{\mu}_a}{\lambda_{\mathsf{SN}}\bar{\gamma}}\right)\frac{\bar{\gamma}}{\tilde{\mu}_a} + \frac{k}{\lambda_{\mathsf{SN}}}\left(C_E + \ln\left(\frac{k\tilde{\mu}_a}{\lambda_{\mathsf{SN}}\bar{\gamma}}\right)\right)\right]\right]\right] \tag{32}$$

A. Determination of the Power Allocation Coefficients and Simulation Settings

In order to select the appropriate PACs for our plots, let us briefly introduce how the PACs have been determined in the literature. Indeed, there are few methods to determine the PACs in NOMA, such as fixed PACs, where the coefficients can be randomly chosen provided that the principle of NOMA is satisfied, i.e., $p_{\mathsf{N}} < p_{\mathsf{F}}$ and $p_{\mathsf{N}} + p_{\mathsf{F}} = 1$ as in [10], [27], [28], or dynamic PACs, where the coefficients can be determined based on the QoS of users, e.g., the target data rate of users [29] or beamforming design [23].

In this paper, we dynamically determine the power allocation coefficients, based on the target data rates of both the cell-center and cell-edge users, as recently done in [29]. For a fair comparison with the non-cooperative NOMA system, the PACs are determined based on the QoS of the cell-center user, i.e., User N, before it performs energy harvesting[4]. The reason of choosing User N is that the SIC process at the cell-center user involves decoding both x_{F} and x_{N}. For the sake of notational convenience, let $\bar{\gamma}_{\mathsf{N}} = \frac{P_{\mathsf{S}}}{\sigma_{\mathsf{aN}}^2 + \sigma_{\mathsf{cN}}^2}$. Thus, the SINR at User N to decode x_{F} in (4) can be now re-expressed as

$$\gamma_{i\mathsf{N}}^{x_{\mathsf{F}}} = \frac{p_{\mathsf{F}}\bar{\gamma}_{\mathsf{N}}|h_{i\mathsf{N}}|^2}{p_{\mathsf{N}}\bar{\gamma}_{\mathsf{N}}|h_{i\mathsf{N}}|^2 + 1}, \tag{33}$$

[4]In [29], only power-splitting was considered so that one could determine PACs independently with ρ. On the other hand, in this paper, we consider a hybrid TS/PS architecture, therefore, it is hard to determine $p_{\mathsf{N}}, p_{\mathsf{F}}$ independently with both α and ρ. For example, we cannot eliminate α from the term $2^{2R_{\mathtt{th},\mathsf{F}}/(1-\alpha)}$.

and the SNR at User N to decode x_{N} in (5) can be rewritten as

$$\gamma_{i\mathsf{N}}^{x_{\mathsf{N}}} = p_{\mathsf{N}}\bar{\gamma}_{\mathsf{N}}|h_{i\mathsf{N}}|^2. \tag{34}$$

The achievable rates for User N to detect the message of User F, x_{F}, can be expressed as

$$R_{\mathsf{N}}^{x_{\mathsf{F}}} = \frac{1}{2}\log_2\left(1 + \gamma_{i\mathsf{N}}^{x_{\mathsf{F}}}\right), \tag{35}$$

and achievable rates for User N to detect its own message, x_{N}, can be expressed as

$$R_{\mathsf{N}}^{x_{\mathsf{N}}} = \frac{1}{2}\log_2\left(1 + \gamma_{i\mathsf{N}}^{x_{\mathsf{N}}}\right). \tag{36}$$

The factor $\frac{1}{2}$ appears in (35) and (36) because α is assumed to be zero. In order to ensure that User N correctly decodes first x_{F} and then x_{N}, we impose that $R_{\mathsf{N}}^{x_{\mathsf{F}}} = R_{\mathtt{th},\mathsf{F}}$ and $R_{\mathsf{N}}^{x_{\mathsf{N}}} = R_{\mathtt{th},\mathsf{N}}$, where $R_{\mathtt{th},\mathsf{F}}$ and $R_{\mathtt{th},\mathsf{N}}$ (bits/s/Hz) denote the target data for decoding x_{F} and x_{N}, respectively. Thus, we have

$$\begin{cases} \frac{1}{2}\log_2\left(1 + \frac{p_{\mathsf{F}}\bar{\gamma}_{\mathsf{N}}|h_{i\mathsf{N}}|^2}{p_{\mathsf{N}}\bar{\gamma}_{\mathsf{N}}|h_{i\mathsf{N}}|^2 + 1}\right) = R_{\mathtt{th},\mathsf{F}}, \\ \frac{1}{2}\log_2\left(1 + p_{\mathsf{N}}\bar{\gamma}_{\mathsf{N}}|h_{i\mathsf{N}}|^2\right) = R_{\mathtt{th},\mathsf{N}}, \\ p_{\mathsf{F}} + p_{\mathsf{N}} = 1, \end{cases} \tag{37}$$

and after some manipulations, (37) can be rewritten as

$$\begin{cases} \left(1 - p_{\mathsf{N}}2^{2R_{\mathtt{th},\mathsf{F}}}\right)\bar{\gamma}_{\mathsf{N}}|h_{i\mathsf{N}}|^2 = 2^{2R_{\mathtt{th},\mathsf{F}}} - 1, \\ p_{\mathsf{N}}\bar{\gamma}_{\mathsf{N}}|h_{i\mathsf{N}}|^2 = 2^{2R_{\mathtt{th},\mathsf{N}}} - 1, \end{cases} \tag{38}$$

and after some algebraic steps, the power allocation coefficients of Users N and F can be determined as

$$\begin{cases} p_{\mathsf{N}} = \dfrac{2^{2R_{\mathrm{th,N}}} - 1}{2^{2R_{\mathrm{th,N}}+2R_{\mathrm{th,F}}} - 1}, \\ p_{\mathsf{F}} = 1 - p_{\mathsf{N}}. \end{cases} \tag{39}$$

In the simulation setting, we assume that the source, Users N and F form a line network, where User N is located on the the direct line between the BS and User F. It is noteworthy that this setting has been a well-known assumption not only in conventional cooperative networks [16], [30] but also in cooperative NOMA networks [9], [28]. There are few reasons of using this setting. For instance, it can be used to discuss the influence of User N's position on the performance of the two-hop cooperative NOMA systems, where User N is moving along the $\mathsf{S} \rightarrow \mathsf{F}$ line as presented in Fig. 6.

In addition, we assume that the path loss exponent equals 3 for all wireless channels, this assumption has been widely adopted in the literature [31]–[33]. It is noteworthy that the path loss exponent for urban microcells varies typically from 2.7 to 3.5 [34].

Unless otherwise stated, the simulation parameters are presented in Table. I.

TABLE I
SIMULATION PARAMETERS

Parameters	Value
Bandwidth	1 MHz
Antenna noise power density, $n_{\mathsf{aN}} = n_{\mathsf{aF}}$	-100 dBm/Hz
Information processing noise power density, $n_{\mathsf{cN}} = n_{\mathsf{cF}}$	-90 dBm/Hz
Target data rates of Users N and F, $R_{\mathrm{th,N}} = R_{\mathrm{th,F}}$	0.1 bits/s/Hz
The distance between S and User F, d_{SF}	10 meters
The distance between S and User N, d_{SN}	3 meters
The distance between User N and User F, d_{NF}	$d_{\mathsf{SF}} - d_{\mathsf{SN}}$
Path-loss exponent, ϵ	3
Path-loss at reference distance, ($\mathcal{L}$ at $d_0 = 1$ m)	-30 dB
Energy conversion efficiency of the EH process, η	0.7

B. Outage Performance

For demonstration purpose, Fig. 3 presents the OP of Scheme I at User F as a function of the fraction of time for EH, α, and the power-splitting ratio, ρ. As can be observed, the results generated using the obtained closed-form approximate expressions for the OP are well corroborated with the simulation results, which validates our developed analysis. Since similar observations are obtained for Schemes II and III, their numerical examples are omitted to present here. The impacts of α and ρ on the OPs will be elaborated next.

In Figs. 4a and 4b, we plot the OPs of three schemes as functions of α and ρ, respectively. As can be seen, α has stronger impact than ρ on the OPs of the proposed schemes. Indeed, for a given value of ρ, OP significantly changes with different values of α. In contrast, for a given value of ρ, such change of OP is not much considerable. It can be explained that from the standpoint of the performance of the cell-edge user, fixing ρ and varying α affects the performance of both the direct and relaying transmissions, while fixing α and varying ρ just affects only the performance of the relaying transmission.

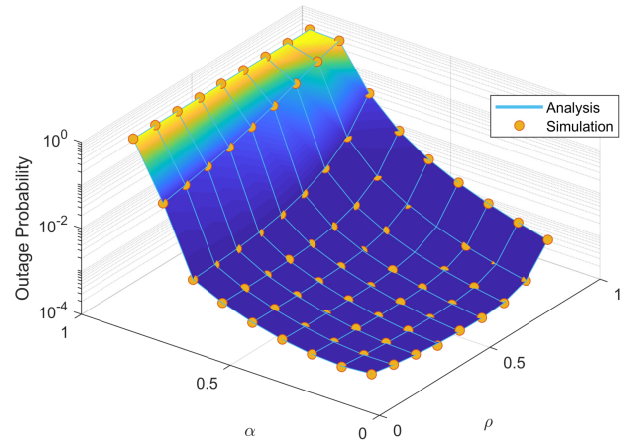

Fig. 3. Outage probability of User F in Scheme I as a function of the fraction of time for EH, α, and the power splitting ratio, ρ, with $K = 3$, $P_{\mathsf{S}} = 30$ dBm.

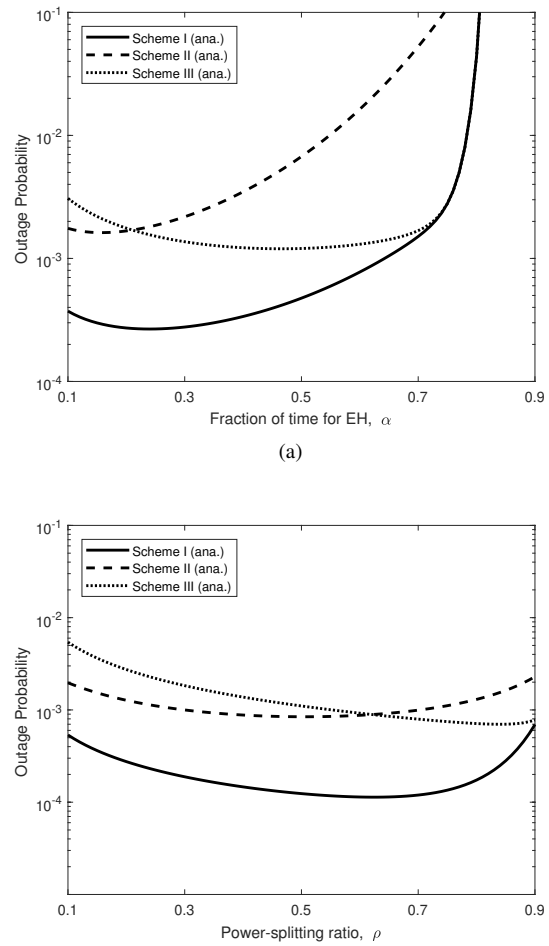

Fig. 4. Outage probability of User F as a function of (a) the fraction of time for EH, α, with $\rho = 0$ and (b) the power-splitting ratio, ρ, with $\alpha = 0$, where in both (a) and (b) we set $K = 3$ and $P_{\mathsf{S}} = 30$ dBm.

Fig. 5 presents a comparison of the OPs of the three proposed schemes while the performance of the random antenna

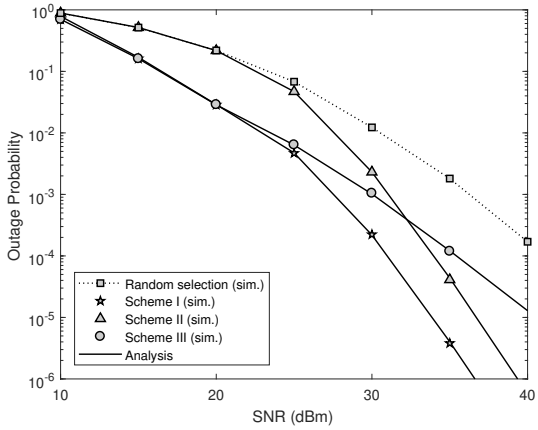

Fig. 5. Comparison of the outage probability of User F in Schemes I, II, III, and the random antenna selection scheme with $K = 3$.

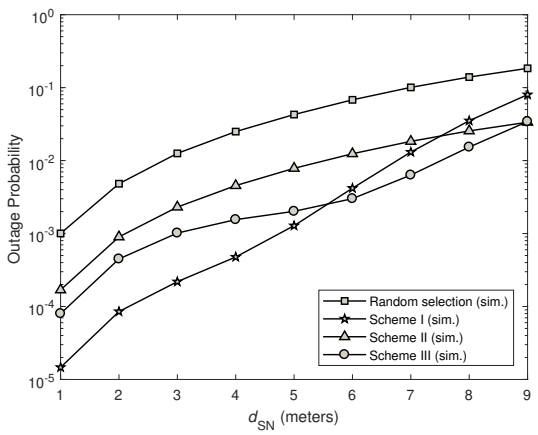

Fig. 6. Outage probability of User F as a function of the distance between S and User N, with $K = 3$, $\alpha = 0.3$, $\rho = 0.3$, and $P_S = 30$ (dBm).

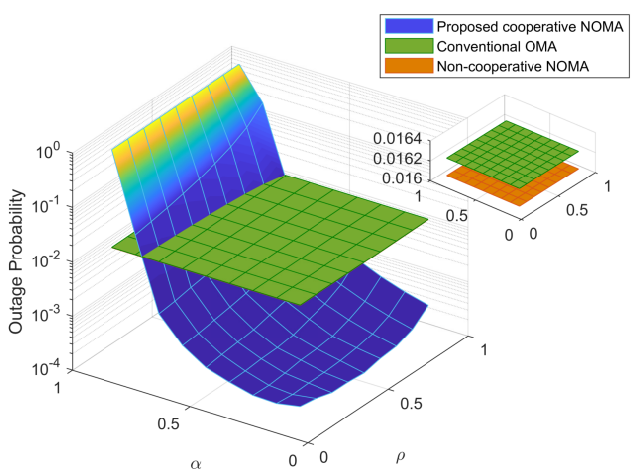

Fig. 7. Cell-edge outage performance comparison of between hybrid SWIPT-based cooperative NOMA, conventional OMA, and non-cooperative NOMA systems, where the random antenna selection is performed with $K = 3$ and $P_S = 30$ dBm.

selection is used as a benchmark. We can see that with the user of the proposed TAS criteria, Schemes I, II, III achieve better outage performance compared to that of the random antenna selection. Among the proposed schemes, Scheme I indeed provides the best outage performance. Additionally, Fig. 5 shows that Scheme II outperforms Criterion III at the high SNR regime, and vice-versa at the low SNR regime.

Next, the distance between the BS and User N, d_{SN}, is discussed more deeply. Fig. 6 plots OP of User F in Schemes I, II, III, and the random antenna selection scheme as a function of d_{SN}. As can be observed, the OP of the TAS schemes increases as d_{SN} increases. One possible reason is that when User N is located more far away from the BS, the amount of harvested energy becomes smaller, and thus, the less power for User N to relay the message of User F. In addition, one of the principles of NOMA is to make use of the difference in the channel conditions of the channels from the BS to the cell-center and cell-edge users, so that the SIC receiver at the cell-center user is able to work better. However, when User N is closer to User F, their channel conditions become more similar, which leads to the decreasing of the performance of NOMA system.

In Fig. 7, we provide a performance comparison between the hybrid SWIPT-based cooperative MISO-NOMA, conventional MISO-OMA, and non-cooperative MISO-NOMA systems. For a fair comparison, the random antenna selection is considered. In the OMA system, the block time T is divided into equal sub-blocks, each one is used to serve one user. In the non-cooperative NOMA system, the source simultaneously serves the two users in the whole block time. As can be seen from Fig. 7, conventional NOMA actually improves the cell-edge performance compared to that of the conventional OMA. Considering the hybrid SWIPT-based cooperative NOMA system, it is obvious that the cell-edge performance becomes worse as α and/or ρ are set too high, i.e., near 1. Since in such scenario, the users do not have sufficient received power to correctly decoded their information.

C. Diversity Gain

We now turn our attention to the achievable diversity gain of the proposed schemes. The physical meaning of the diversity gain can be explained as follows. In wireless communications, diversity technique uses the fact that independent users have a low probability of experiencing deep effects of fading simultaneously [24]. Diversity gain is used as a performance metric of a diversity technique. The diversity gain, D, is mathematically formulated as [35]

$$D = -\lim_{\bar{\gamma}\to\infty} \frac{\log P_e(\bar{\gamma})}{\log \bar{\gamma}}, \tag{40}$$

where $\bar{\gamma}$ denotes SNR and P_e denotes bit error rate as a function of SNR. When the outage probability P_{out} is considered instead of the bit error rate, the diversity gain, D, can be written as [11]

$$D = -\lim_{\bar{\gamma}\to\infty} \frac{\log P_{\text{out}}(\bar{\gamma})}{\log \bar{\gamma}}. \tag{41}$$

Consequently, as the slope of an OP curve of a scheme is parallel to the curve that is proportional to $\frac{1}{\bar{\gamma}^D}$ at high SNR

regime, we can conclude that the diversity gain achieved by that scheme is D [28].

Fig. 8 shows the diversity gain achieved by three proposed schemes. Recall that diversity behavior can be indicated by the slope at the high SNR regime of the outage performance curve [8], [28]. As shown in Fig. 8, the slope of the performance curves of User F in Schemes I and II are parallel to the lines for $\frac{1}{\bar{\gamma}^{K+1}} = \frac{1}{\bar{\gamma}^4}$, while that of User F in Scheme III is parallel to the line for $\frac{1}{\bar{\gamma}^2}$.

Similar trends are observed in Fig. 9, where we plot the OPs of User F with different numbers of antennas. As can be seen, for a given number of antenna, the slope of the performance curves of Schemes I and II are parallel, which confirms that the diversity gains achieved by Schemes I and II are the same. In addition, by substituting the asymptotic OP of Scheme II into (41), and relying on the fact that $\lim_{x\to\infty} \log\left[c_1/x^{K+2} + (c_2/x^{K+1})\ln\left(\frac{c_3}{x}\right)\right]/\log x = -K-1$, where c_1, c_2, c_3 are constants, it can be concluded that the diversity gain achieved by Scheme II at User F is $K+1$.[5]

On the other hand, the slopes of the performance curves of Scheme III are always parallel to each other irrespective of the number of antennas, K. Thus, the diversity gain achieved by Scheme III does not increase as K increases. By carrying out similar diversity analysis as done with Scheme II, we can conclude that the diversity gain achieved by scheme III at the cell-edge user is always 2.

Finally, in order to discuss the complexity of the proposed schemes, we introduce a definition for the complexity order. Specifically, the complexity order of a scheme will be defined in terms of the number of channels that require CSI estimation. Its unit concerns to the number of channels. Note that the BS conducts CSI estimation of the considered channels in order to select an appropriate antenna before transmitting data.

TABLE II
ACHIEVABLE DIVERSITY GAINS AND COMPLEXITY ORDER OF THE PROPOSED SCHEMES

Schemes	Achievable diversity gain	Complexity order
Scheme I	$K+1$	$2K+1$
Scheme II	$K+1$	K
Scheme III	2	K

The achievable diversity gain and complexity order of each scheme are summarized in Table. II. As can be observed, the TAS criterion in Scheme I requires the highest complexity order, i.e., $2K+1$. While Schemes II and III needs the same number of CSI estimation and is indeed lower than that required by Scheme I, i.e., K.

VI. CONCLUSIONS

In this paper, aiming to improve the performance of cell-edge users in two-user MISO-NOMA cooperative downlink transmissions, we have proposed three cooperative schemes where the SWIPT is employed at the cell-center user to power the DF relaying operation. Additionally, each scheme adopts different TAS criterion to provide flexible choices for system designing. Considering the hybrid TS/PS SWIPT architecture, we have investigated the performance of the cell-edge user in terms of outage probability and diversity gain. More specifically, the tight closed-form approximate expressions for the OPs and the corresponding asymptotic OPs achieved by the three schemes have been derived and validated through Monte-Carlo simulations. Numerical results have showed that the proposed schemes provide essential cell-edge performance improvements compared to the conventional OMA and non-cooperative NOMA systems. Particularly, Scheme I obtains the optimal outage performance with the diversity gain of $K+1$, where K is the number of antennas. Scheme II achieves lower performance compared to that of Scheme I but the same diversity gain. Scheme III outperforms Scheme II at low SNR regime and vice versa at high SNR regime while it gives the lowest diversity gain, i.e., always 2.

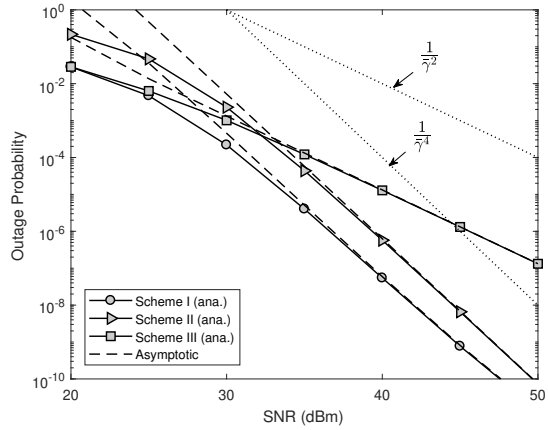

Fig. 8. Illustrations of the diversity gain achieved by the proposed schemes with $K=3$, $\alpha=0.3$, and $\rho=0.3$.

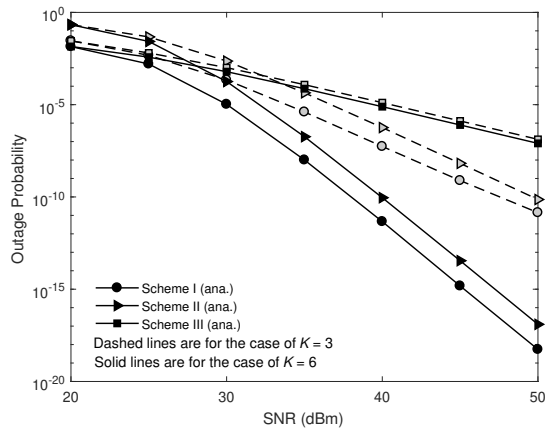

Fig. 9. Illustrations of the diversity gain achieved by the proposed schemes under different numbers of antennas with $\alpha=0.3$ and $\rho=0.3$.

[5]Note that this analytical proof cannot be applied for Scheme I since when substituting the asymptotic OP of Scheme I into (41), the limit becomes intricate and cannot be obtained.

Appendix A
Proof of Theorem 1

From (19), the OP of User F in Scheme I can be written as

$$P_{\text{out,F}}^{(\text{I})} = \Pr\left(\max_{1\le i\le K} \min\left\{\frac{a_1 X_i}{a_2 X_i + 1}, \max\left\{\frac{b_1 Z_i}{b_2 Z_i + 1}, cX_i Y\right\}\right\} < \gamma_2\right). \quad (42)$$

As we can observe, the events of the probability in (42) are not mutually exclusive because they include the same random variable Y. Therefore, conditioning on $Y = y$, $P_{\text{out,F}}^{(\text{I})}$ can be re-expressed as

$$P_{\text{out,F}}^{(\text{I})} = \int_0^\infty \Pr\left(\max_{1\le i\le K} \min\left\{\frac{a_1 X_i}{a_2 X_i + 1}, \max\left\{\frac{b_1 Z_i}{b_2 Z_i + 1}, cX_i y\right\}\right\} < \gamma_2\right) f_Y(y)dy. \quad (43)$$

Next, conditioning on $X_i = x$, $P_{\text{out,F}}^{(\text{I})}$ can be further expressed as

$$\begin{aligned} P_{\text{out,F}}^{(\text{I})} &= \int_0^\infty \prod_{i=1}^K \int_0^\infty \Pr\left(\min\left\{\frac{a_1 x}{a_2 x + 1}, \max\left\{\frac{b_1 Z_i}{b_2 Z_i + 1}, cxy\right\}\right\} < \gamma_2\right) f_X(x)dx f_Y(y)dy \\ &= \int_0^\infty \prod_{i=1}^K \left[1 - \int_0^\infty \Pr\left(\frac{a_1 x}{a_2 x + 1} \ge \gamma_2\right) \times \Pr\left(\max\left\{\frac{b_1 Z_i}{b_2 Z_i + 1}, cxy\right\} \ge \gamma_2\right) f_X(x)dx\right] \times f_Y(y)dy. \end{aligned} \quad (44)$$

As can be observed, if $\gamma_2 \ge \theta$ then $P_{\text{out,F}}^{(\text{I})} = 1$. For the case $\gamma_2 < \theta$, let $\mu_a = \frac{\gamma_2}{a_1 - a_2\gamma_2}$, $P_{\text{out,F}}^{(\text{I})}$ can be expressed as

$$\begin{aligned} P_{\text{out,F}}^{(\text{I})} &= \int_0^\infty \prod_{i=1}^K \left[1 - \int_{\mu_a}^\infty \left[1 - \Pr\left(\max\left\{\frac{b_1 Z_i}{b_2 Z_i + 1}, cxy\right\} < \gamma_2\right)\right] f_X(x)dx\right] f_Y(y)dy \\ &= \int_0^\infty \prod_{i=1}^K \left[1 - e^{-\frac{\mu_a}{\lambda_X}} + \int_{\mu_a}^\infty \Pr\left(\frac{b_1 Z_i}{b_2 Z_i + 1} < \gamma_2\right) \times \Pr\left(x < \frac{\gamma_2}{cy}\right) f_x(x)dx\right] f_Y(y)dy. \end{aligned} \quad (45)$$

Since $\mu_a \le x < \frac{\gamma_2}{cy}$, we first consider the case $y < \frac{\gamma_2}{c\mu_a}$, let $\mu_b = \frac{\gamma_2}{b_1 - b_2\gamma_2}$, and after some algebraic steps, $P_{\text{out,F}}^{(\text{I})}$ can be written as

$$\begin{aligned} P_{\text{out,F}}^{(\text{I})} = \Xi_1 &\triangleq \int_0^{\frac{\gamma_2}{c\mu_a}} \prod_{i=1}^K \left[1 - e^{-\frac{\mu_a}{\lambda_X}} + \int_{\mu_a}^{\frac{\gamma_2}{cy}} \left(1 - e^{-\frac{\mu_b}{\lambda_Z}}\right) \frac{1}{\lambda_X} e^{-\frac{x}{\lambda_X}} dx\right] f_Y(y)dy \\ &= \int_0^{\frac{\gamma_2}{c\mu_a}} \left[1 - e^{-\frac{\mu_a}{\lambda_X}} + \left(1 - e^{-\frac{\mu_b}{\lambda_Z}}\right) \times \left(e^{-\frac{\mu_a}{\lambda_X}} - e^{-\frac{\gamma_2}{\lambda_X cy}}\right)\right]^K \frac{1}{\lambda_Y} e^{-\frac{y}{\lambda_Y}} dy. \end{aligned} \quad (46)$$

In order to further simplify the above integral, we rely on the trinomial coefficient, i.e., $(\alpha + \beta + \gamma)^K = \sum_{i,j,k}^{K=i+j+k} \binom{K}{i,j,k} \alpha^i \beta^j \gamma^k$ [36]. Thus, Ξ_1 can be expressed as

$$\Xi_1 = \sum_{i,j,k}^{K=i+j+k} \binom{K}{i,j,k} \left(1 - e^{-\frac{\mu_a}{\lambda_X} - \frac{\mu_b}{\lambda_Z}}\right)^i (-1)^j e^{-\frac{k\mu_b}{\lambda_Z}} \times \frac{1}{\lambda_Y} \underbrace{\int_0^{\frac{t}{c\mu_a}} e^{-\frac{y}{\lambda_Y} - \frac{(j+k)\gamma_2}{\lambda_X cy}} dy}_{I_1}. \quad (47)$$

The integral I_1 in (47) can be expressed as

$$I_1 = \int_0^\infty e^{-\frac{y}{\lambda_Y} - \frac{(j+k)\gamma_2}{\lambda_X cy}} dy - \int_{\frac{\gamma_2}{c\mu_a}}^\infty e^{-\frac{y}{\lambda_Y} - \frac{(j+k)\gamma_2}{\lambda_X cy}} dy. \quad (48)$$

By making use of the relationship $\int_0^\infty e^{-\chi x - \frac{\xi}{4x}} = \sqrt{\frac{\xi}{\chi}} K_1(\sqrt{\xi\chi})$ [25, Eq. (3.324.1)], where $K_1(\cdot)$ is the first-order modified Bessel function of the second kind [25, Eq. (8.407.1)], and relying on a result in our previous work [37], i.e. $\int_\mu^\infty e^{-\chi x - \frac{\xi}{x}} dx \approx \Omega(\mu, \chi, \xi)$, where $\Omega(\mu, \chi, \xi)$ is defined in (21), I_1 can be obtained as

$$I_1 = \sqrt{\frac{4\xi}{\chi}} K_1(\sqrt{4\xi\chi}) - \Omega\left(\frac{\gamma_2}{c\mu_a}, \frac{1}{\lambda_{\text{SN}}}, \frac{(j+k)\gamma_2}{\lambda_X c}\right). \quad (49)$$

Next, considering the case $y \ge \frac{\gamma_2}{c\mu_a}$, $P_{\text{out,F}}^{(\text{I})}$ can be obtained as

$$\begin{aligned} P_{\text{out,F}}^{(\text{I})} = \Xi_2 &= \int_{\frac{\gamma_2}{c\mu_a}}^\infty \prod_{i=1}^K \left(1 - e^{-\frac{\mu_a}{\lambda_X}}\right) f_Y(y)dy \\ &= \int_{\frac{\gamma_2}{c\mu_a}}^\infty \left(1 - e^{-\frac{\mu_a}{\lambda_X}}\right)^K \frac{1}{\lambda_Y} e^{-\frac{y}{\lambda_Y}} \\ &= \left(1 - e^{-\frac{\mu_a}{\lambda_X}}\right)^K e^{-\frac{\gamma_2}{\lambda_Y c\mu_a}}. \end{aligned} \quad (50)$$

Combining Ξ_1 and Ξ_2, the OP of User F in Scheme I can be obtained as presented in (20). This completes the proof of Theorem 1.

Appendix B
Proof of Theorem 2

From (22), the OP of User F can be expressed as

$$P_{\text{out,F}}^{(\text{II})} = \underbrace{\Pr(\gamma_{i^*\text{N}}^{x_\text{F}} < \gamma_2, \gamma_{i^*\text{F}} < \gamma_2)}_{\Lambda_1} + \underbrace{\Pr(\gamma_{i^*\text{N}}^{x_\text{F}} \ge \gamma_2, \max\{\gamma_{i^*\text{F}}, \gamma_{\text{NF}}\} < \gamma_2)}_{\Lambda_2}, \quad (51)$$

where the superscript (II) represents a notation for Scheme II.

From (4), (7) and (17), Λ_1 in (51) can be rewritten as

$$\Lambda_1 = \Pr\left(\frac{a_1|h_{i^*\mathsf{N}}|^2}{a_2|h_{i^*\mathsf{N}}|^2+1} < \gamma_2, \frac{b_1|h_{i^*\mathsf{F}}|^2}{b_2|h_{i^*\mathsf{F}}|^2+1} < \gamma_2\right), \quad (52)$$

Since Users N and F are assumed to be independent, Λ_1 can be further expressed as

$$\Lambda_1 = \underbrace{\Pr\left(|h_{i^*\mathsf{N}}|^2 < \frac{\gamma_2}{a_1 - a_2\gamma_2}\right)}_{\Lambda_{1A}} \underbrace{\Pr\left(|h_{i^*\mathsf{F}}|^2 < \frac{\gamma_2}{b_1 - b_2\gamma_2}\right)}_{\Lambda_{1B}}, \quad (53)$$

if $\gamma_2 < \theta$, otherwise, $\Lambda_1 = 1$. It can be explained that when $a_1 - a_2\gamma_2 \leq 0$ and $b_1 - b_2\gamma_2 \leq 0$, respectively, which turns out $\gamma_2 \geq \frac{p_\mathsf{F}}{p_\mathsf{N}}$, Λ_{1A} and Λ_{1B} always equal to 1.

In Scheme II, the antenna that has the largest channel gain of the channel from the BS to User F will be selected. In this case, the statistical characteristics of the channel from the selected antenna at the BS to User N can be determined as follows. Using the total probability theory [38], the cumulative distribution function (CDF) of $|h_{i^*\mathsf{N}}|^2$ can be expressed as

$$\begin{aligned} F_{|h_{i^*\mathsf{N}}|^2}(x) &= \Pr(|h_{i^*\mathsf{N}}|^2 < x) \\ &= \sum_{i=1}^{K} \Pr(i^* = i) \Pr(|h_{i\mathsf{N}}|^2 < x), \end{aligned} \quad (54)$$

where $\Pr(i^* = i)$ in this case, i.e., Scheme II, can be obtained as

$$\begin{aligned} \Pr(i^* = i) &= \Pr\left(\bigcap_{j=1, j\neq i}^{K} |h_{j\mathsf{F}}|^2 < |h_{i\mathsf{F}}|^2\right) \\ &= \int_0^\infty \prod_{j=1, j\neq i}^{K} \left[1 - \Pr(|h_{j\mathsf{F}}|^2 < x)\right] f_{|h_{i\mathsf{F}}|^2}(x)dx \\ &= 1/K. \end{aligned} \quad (55)$$

Consequently, the CDF and the probability density function (PDF) of $|h_{i^*\mathsf{N}}|^2$ can be, respectively, written by

$$F_{|h_{i^*\mathsf{N}}|^2}(x) = 1 - e^{-\frac{x}{\lambda_\mathsf{SN}}}, \quad f_{|h_{i^*\mathsf{N}}|^2}(x) = \frac{1}{\lambda_\mathsf{SN}} e^{-\frac{x}{\lambda_\mathsf{SN}}}. \quad (56)$$

Thus, Λ_{1A} can be obtained as

$$\Lambda_{1A} = 1 - e^{-\frac{\gamma_2}{\lambda_\mathsf{SN}(a_1 - a_2\gamma_2)}}, \quad (57)$$

if $\gamma_2 < \frac{p_\mathsf{F}}{p_\mathsf{N}}$, otherwise, $\Lambda_{1A} = 1$.

Following the TAS criterion adopted in Scheme II, with the aid of order statistics [38], and [25, Eq. (1.111)], the CDF and PDF of $|h_{i^*\mathsf{F}}|^2$ can be written, respectively, as

$$F_{|h_{i^*\mathsf{F}}|^2}(y) = \sum_{k=0}^{K} \binom{K}{k} (-1)^k e^{-\frac{kx}{\lambda_\mathsf{SF}}}, \quad (58)$$

$$f_{|h_{i^*\mathsf{F}}|^2}(x) = \sum_{k=1}^{K} \binom{K}{k} (-1)^{k+1} \frac{k}{\lambda_X} e^{-\frac{kx}{\lambda_\mathsf{SF}}}. \quad (59)$$

Thus, Λ_{1B} can be obtained as

$$\Lambda_{1B} = \sum_{k=0}^{K} \binom{K}{k} (-1)^k e^{-\frac{k\gamma_2}{\lambda_\mathsf{SF}(b_1 - b_2\gamma_2)}}, \quad (60)$$

if $\gamma_2 < \theta$, otherwise, $\Lambda_{1B} = 1$.

Next, since Users N and F are assumed to be independent, Λ_2 in (51) can be rewritten as

$$\Lambda_2 = \underbrace{\Pr(\gamma_{i^*\mathsf{F}} < \gamma_2)}_{\Lambda_{2A}} \underbrace{\Pr(\gamma_{i^*\mathsf{N}}^{x_\mathsf{F}} \geq \gamma_2, \gamma_\mathsf{NF} < \gamma_2)}_{\Lambda_{2B}}, \quad (61)$$

where $\Lambda_{2A} = \Lambda_{1B}$, and Λ_{2B} can be re-expressed as

$$\Lambda_{2B} = \Pr\left(\frac{a_1|h_{i^*\mathsf{N}}|^2}{a_2|h_{i^*\mathsf{N}}|^2+1} \geq \gamma_2, c|h_{i^*\mathsf{N}}|^2|h_\mathsf{NF}|^2 < \gamma_2\right). \quad (62)$$

Note that $|h_\mathsf{NF}|^2$ follows exponential distribution with mean λ_NF. Conditioning on $|h_{i^*\mathsf{N}}|^2 = x$, (62) becomes

$$\Lambda_{2B} = \int_0^\infty \Pr\left(x \geq \mu_a, |h_\mathsf{NF}|^2 < \frac{\gamma_2}{cx}\right) f_{|h_{i^*\mathsf{N}}|^2}(x)dx, \quad (63)$$

where $\mu_a \triangleq \gamma_2/(a_1 - a_2\gamma_2)$, and can be further expressed as

$$\Lambda_{2B} = \int_{\mu_a}^\infty \Pr\left(|h_\mathsf{NF}|^2 < \frac{\gamma_2}{cx}\right) f_{|h_{i^*\mathsf{N}}|^2}(x)dx \quad (64)$$

$$= e^{-\frac{\mu_a}{\lambda_\mathsf{SN}}} - \frac{1}{\lambda_\mathsf{SN}} \underbrace{\int_{\mu_a}^\infty e^{-\frac{\gamma_2}{\lambda_\mathsf{NF}cx} - \frac{x}{\lambda_\mathsf{SN}}} dx}_{I_2}. \quad (65)$$

I_2 can be obtained as done in Appendix A. Thus, Λ_{2B} can be obtained as

$$\Lambda_{2B} = e^{-\frac{\gamma_2}{\lambda_\mathsf{SN}(a_1 - a_2\gamma_2)}} - \frac{1}{\lambda_\mathsf{SN}} \Omega\left(\frac{\gamma_2}{a_1 - a_2\gamma_2}, \frac{1}{\lambda_\mathsf{SN}}, \frac{\gamma_2}{\lambda_\mathsf{NF}c}\right). \quad (66)$$

By substituting $\Lambda_{1A}, \Lambda_{1B}$ and Λ_{2B} into (51), the closed-form approximate expression for the OP of Use F in Scheme II can be obtained. This completes the proof of Theorem 2.

References

[1] Y. Saito, Y. Kishiyama, A. Benjebbour, T. Nakamura, A. Li, and K. Higuchi, "Non-orthogonal multiple access (NOMA) for cellular future radio access," in *Proc. IEEE Veh. Technol. Conf. (VTC Spring)*, Dresden, Germany, Jun. 2013, pp. 1–5.

[2] NTT DOCOMO, "5G radio access: requirements, concept and technologies," *White Paper*, Jun. 2014.

[3] Z. Ding, C. Zhong, D. W. K. Ng, M. Peng, H. A. Suraweera, R. Schober, and H. V. Poor, "Application of smart antenna technologies in simultaneous wireless information and power transfer," *IEEE Commun. Mag.*, vol. 53, no. 4, pp. 86–93, Apr. 2015.

[4] T. Shimojo, A. Umesh, D. Fujishima, and A. Minokuchi, "Special articles on 5G technologies toward 2020 deployment," *NTT DOCOMO Tech. J.*, vol. 17, no. 4, pp. 50–59, 2016.

[5] L. Dai, B. Wang, Y. Yuan, S. Han, C. l. I, and Z. Wang, "Non-orthogonal multiple access for 5G: solutions, challenges, opportunities, and future research trends," *IEEE Commun. Mag.*, vol. 53, no. 9, pp. 74–81, Sep. 2015.

[6] J. Choi, "Minimum power multicast beamforming with superposition coding for multiresolution broadcast and application to NOMA systems," *IEEE Trans. Commun.*, vol. 63, no. 3, pp. 791–800, Mar. 2015.

[7] V.-D. Nguyen, H. D. Tuan, T. Q. Duong, H. V. Poor, and O.-S. Shin, "Precoder design for signal superposition in MIMO-NOMA multicell networks," *IEEE J. Select. Areas Commun.*, 2017, doi: 10.1109/JSAC.2017.2726007.

[8] Z. Ding, M. Peng, and H. V. Poor, "Cooperative non-orthogonal multiple access in 5G systems," *IEEE Commun. Lett.*, vol. 19, no. 8, pp. 1462–1465, Aug. 2015.

[9] J. So and Y. Sung, "Improving non-orthogonal multiple access by forming relaying broadcast channels," *IEEE Commun. Lett.*, vol. 20, no. 9, pp. 1816–1819, Sep. 2016.

[10] N. T. Do, D. B. da Costa, T. Q. Duong, and B. An, "A BNBF user selection scheme for NOMA-based cooperative relaying systems with SWIPT," *IEEE Commun. Lett.*, vol. 21, no. 3, pp. 664–667, Mar. 2017.

[11] Y. Liu, Z. Ding, M. Elkashlan, and H. V. Poor, "Cooperative non-orthogonal multiple access with simultaneous wireless information and power transfer," *IEEE J. Select. Areas Commun.*, vol. 34, no. 4, pp. 938–953, Apr. 2016.

[12] P. D. Diamantoulakis, K. N. Pappi, G. K. Karagiannidis, H. Xing, and A. Nallanathan, "Joint downlink/uplink design for wireless powered networks with interference," *IEEE Access*, vol. 5, pp. 1534–1547, Jan. 2017.

[13] Z. Chen, Z. Ding, X. Dai, and G. K. Karagiannidis, "On the application of quasi-degradation to MISO-NOMA downlink," *IEEE Trans. Signal Processing*, vol. 64, no. 23, pp. 6174–6189, Dec. 2016.

[14] Z. Chen, Z. Ding, P. Xu, and X. Dai, "Optimal precoding for a QoS optimization problem in two-user MISO-NOMA downlink," *IEEE Commun. Lett.*, vol. 20, no. 6, pp. 1263–1266, Jun. 2016.

[15] P. L. Yeoh, M. Elkashlan, N. Yang, D. B. da Costa, and T. Q. Duong, "Unified analysis of transmit antenna selection in MIMO multirelay networks," *IEEE Trans. Veh. Tech.*, vol. 62, no. 2, pp. 933–939, Feb. 2013.

[16] D. C. Gonzalez, D. B. da Costa, and J. C. S. S. Filho, "Distributed TAS/MRC and TAS/SC schemes for fixed-gain AF systems with multiantenna relay: outage performance," *IEEE Trans. Wireless Commun.*, vol. 15, no. 6, pp. 4380–4392, Jun. 2016.

[17] G. Pan, H. Lei, Y. Deng, L. Fan, J. Yang, Y. Chen, and Z. Ding, "On secrecy performance of MISO SWIPT systems with TAS and imperfect CSI," *IEEE Trans. Commun.*, vol. 64, no. 9, pp. 3831–3843, Sep. 2016.

[18] X. Liu and X. Wang, "Efficient antenna selection and user scheduling in 5G massive MIMO-NOMA system," in *2016 IEEE 83rd Veh. Tech. Conf. (VTC Spring)*, Nanjing, China, May 2016, pp. 1–5.

[19] W. Han, J. Ge, and J. Men, "Performance analysis for NOMA energy harvesting relaying networks with transmit antenna selection and maximal-ratio combining over Nakagami-m fading," *IET Commun.*, Sep. 2016.

[20] H. Liu, K. J. Kim, K. S. Kwak, and H. V. Poor, "Power splitting-based SWIPT with decode-and-forward full-duplex relaying," *IEEE Trans. Wireless Commun.*, vol. 15, no. 11, pp. 7561–7577, Nov. 2016.

[21] S. Atapattu and J. Evans, "Optimal energy harvesting protocols for wireless relay networks," *IEEE Trans. Wireless Commun.*, vol. 15, no. 8, pp. 5789–5803, Aug. 2016.

[22] R. Tao, A. Salem, and K. A. Hamdi, "Adaptive relaying protocol for wireless power transfer and information processing," *IEEE Commun. Lett.*, vol. 20, no. 10, pp. 2027–2030, Oct. 2016.

[23] D. Zhang, Y. Liu, Z. Ding, Z. Zhou, A. Nallanathan, and T. Sato, "Performance analysis of non-regenerative massive-MIMO-NOMA relay systems for 5G," *IEEE Trans. Commun.*, vol. 65, no. 11, pp. 4777–4790, Nov. 2017.

[24] A. Goldsmith, *Wireless Communications*. Cambridge, U.K.: Cambridge Univ. Press, 2005.

[25] I. S. Gradshteyn and I. M. Ryzhik, *Tables of Integrals, Series, and Products*, 7th ed. New York, NY, USA: Academic Press, 2007.

[26] M. Ju, K. M. Kang, K. S. Hwang, and C. Jeong, "Maximum transmission rate of PSR/TSR protocols in wireless energy harvesting DF-based relay networks," *IEEE J. Select. Areas Commun.*, vol. 33, no. 12, pp. 2701–2717, Dec. 2015.

[27] J. B. Kim and I. H. Lee, "Capacity analysis of cooperative relaying systems using non-orthogonal multiple access," *IEEE Commun. Lett.*, vol. 19, no. 11, pp. 1949–1952, Nov. 2015.

[28] Z. Ding, F. Adachi, and H. V. Poor, "The application of MIMO to non-orthogonal multiple access," *IEEE Trans. Wireless Commun.*, vol. 15, no. 1, pp. 537–552, Jan. 2016.

[29] Z. Yang, Z. Ding, P. Fan, and N. Al-Dhahir, "The impact of power allocation on cooperative non-orthogonal multiple access networks with SWIPT," *IEEE Trans. Wireless Commun.*, vol. 16, no. 7, pp. 4332–4343, Jul. 2017.

[30] V. N. Q. Bao, T. Q. Duong, and C. Tellambura, "On the performance of cognitive underlay multihop networks with imperfect channel state information," *IEEE Trans. Commun.*, vol. 61, no. 12, pp. 4864–4873, Dec. 2013.

[31] Z. Ding, Z. Yang, P. Fan, and H. V. Poor, "On the performance of non-orthogonal multiple access in 5G systems with randomly deployed users," *IEEE Signal Processing Lett.*, vol. 21, no. 12, pp. 1501–1505, Dec. 2014.

[32] J. Choi, "Effective capacity of NOMA and a suboptimal power control policy with delay QoS," *IEEE Trans. Commun.*, vol. 65, no. 4, pp. 1849–1858, Apr. 2017.

[33] Y. Zhang, H. M. Wang, Q. Yang, and Z. Ding, "Secrecy sum rate maximization in non-orthogonal multiple cccess," *IEEE Commun. Lett.*, vol. 20, no. 5, pp. 930–933, May 2016.

[34] A. Goldsmith, *Wireless Communications*. New York, NY, USA: Cambridge University Press, 2005.

[35] L. Zheng and D. N. C. Tse, "Diversity and multiplexing: a fundamental tradeoff in multiple-antenna channels," *IEEE Trans. Inf. Theory*, vol. 49, no. 5, pp. 1073–1096, May 2003.

[36] T. Koshy, *Discrete Mathematics with Applications*. Burlington, VT, USA: Academic Press, 2004.

[37] N. T. Do, D. B. da Costa, T. Q. Duong, V. N. Q. Bao, and B. An, "Exploiting direct links in multiuser multirelay SWIPT cooperative networks with opportunistic scheduling," *IEEE Trans. Wireless Commun.*, vol. 16, no. 8, pp. 5410–5427, Aug. 2017.

[38] A. Papoulis and S. U. Pillai, *Probability, random variables, and stochastic processes*, 4th ed. New York, NY, USA: McGraw-Hill, 2002.

Tri Nhu Do (S'16) was born and raised in Da Nang, Vietnam. He received the B.S. degree in electronics and telecommunications engineering from the Posts and Telecommunications Institute of Technology, Vietnam, in 2012, and the M.S. degree in electronics and computer engineering from Hongik University, Sejong Campus, South Korea, in 2015. He is currently pursuing the Ph.D. degree in electronics and computer engineering with Hongik University.

His main research topics are wireless communications and cooperative relaying transmissions.

Daniel Benevides da Costa (S'04–M'08–SM'14) was born in Fortaleza, Ceará, Brazil, in 1981. He received the B.Sc. degree in Telecommunications from the Military Institute of Engineering (IME), Rio de Janeiro, Brazil, in 2003, and the M.Sc. and Ph.D. degrees in Electrical Engineering, Area: Telecommunications, from the University of Campinas, SP, Brazil, in 2006 and 2008, respectively. His Ph.D thesis was awarded the Best Ph.D. Thesis in Electrical Engineering by the Brazilian Ministry of Education (CAPES) at the 2009 CAPES Thesis Contest. From 2008 to 2009, he was a Postdoctoral Research Fellow with INRS-EMT, University of Quebec, Montreal, QC, Canada. Since 2010, he has been with the Federal University of Ceará, where he is currently an Assistant Professor.

Prof. da Costa is currently Editor of the IEEE COMMUNICATIONS SURVEYS AND TUTORIALS, IEEE ACCESS, IEEE TRANSACTIONS ON COMMUNICATIONS, IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, EURASIP JOURNAL ON WIRELESS COMMUNICATIONS AND NETWORKING, and KSII TRANSACTIONS ON INTERNET AND INFORMATION SYSTEMS. He has also served as Associate Technical Editor of the IEEE COMMUNICATIONS MAGAZINE. From 2012 to 2017, he was Editor of the IEEE COMMUNICATIONS LETTERS. He has served as Guest Editor of several Journal Special Issues. He has been involved on the Organizing Committee of several conferences. He is currently the Latin American Chapters Coordinator of the IEEE Vehicular Technology Society. Also, he acts as a Scientific Consultant of the National Council of Scientific and Technological Development (CNPq), Brazil and he is a Productivity Research Fellow of CNPq. From 2012 to 2017, he was Member of the Advisory Board of the Ceará Council of Scientific and Technological Development (FUNCAP), Area: Telecommunications.

Prof. da Costa is the recipient of three conference paper awards. He received the Exemplary Reviewer Certificate of the IEEE WIRELESS COMMUNICATIONS LETTERS in 2013, the Exemplary Reviewer Certificate of the IEEE COMMUNICATIONS LETTERS in 2016, the Certificate of Appreciation of Top Associate Editor for outstanding contributions to IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY in 2013, 2015 and 2016, and the Exemplary Editor Award of IEEE COMMUNICATIONS LETTERS in 2016. He is a Distinguished Lecturer of the IEEE Vehicular Technology Society. He is a Senior Member of IEEE, Member of IEEE Communications Society and IEEE Vehicular Technology Society.

Trung Q. Duong (S'05, M'12, SM'13) received his Ph.D. degree in Telecommunications Systems from Blekinge Institute of Technology (BTH), Sweden in 2012. Since 2013, he has joined Queen's University Belfast, UK as a Lecturer (Assistant Professor). His current research interests include small-cell networks, ultra-dense networks, physical layer security, energy-harvesting communications, massive MIMO. He is the author or co-author of more than 280 technical papers published in scientific journals (156 articles) and presented at international conferences (125 papers).

Dr. Duong currently serves as an Editor for the IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS, IEEE TRANSACTIONS ON COMMUNICATIONS, IET COMMUNICATIONS, and a Senior Editor for IEEE COMMUNICATIONS LETTERS. He was awarded the Best Paper Award at the IEEE Vehicular Technology Conference (VTC-Spring) in 2013, IEEE International Conference on Communications (ICC) 2014, and IEEE Global Communications Conference (GLOBECOM) 2016. He is the recipient of prestigious Royal Academy of Engineering Research Fellowship (2016-2021) and has won a prestigious Newton Prize 2017.

Beongku An received the M.S. degree in electrical engineering from the New York University (Polytechnic), NY, USA, in 1996 and Ph.D. degree from New Jersey Institute of Technology (NJIT), NJ, USA, in 2002, BS degree in electronic engineering from Kyungpook National university, Korea, in 1988, respectively. After graduation, he joined the Faculty of the Department of Computer and Information Communications Engineering, Hongik University in Korea, where he is currently a Professor. From 1989 to 1993, he was a senior researcher in RIST, Pohang, Korea. He also was lecturer and RA in NJIT from 1997 to 2002. He was a president of IEIE Computer Society (The Institute of Electronics and Information Engineers, Computer Society) in 2012. From 2013, he also works as a General Chair in the International Conference, ICGHIT (International Conference on Green and Human Information Technology).

His current research interests include mobile wireless networks and communications such as ad-hoc networks, sensor networks, wireless internet, cognitive radio networks, ubiquitous networks, cellular networks, and IoT. In particular, he is interested in cooperative routing, multicast routing, energy harvesting, physical layer security, visible light communication (VLC), cross-layer technology, mobile cloud computing. Professor An was listed in Marquis Who's Who in Science and Engineering, and Marquis Who's Who in the World, respectively.